

STATE QUOTIENTS AND AGENT-PERMUTATION SHARING FOR MARKOV POTENTIAL GAMES

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Paper under double-blind review

1 PROBLEM SETUP

For a finite set $\mathcal{X}$, let $|\mathcal{X}|$ denote its cardinality and $\Delta(\mathcal{X})$ the probability simplex over it. We write $[n] := \{1, \dots, n\}$.

1.1 FINITE DISCOUNTED MARKOV GAMES

A finite discounted Markov game is

$$M = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, P, \vec{r}, \mu, \gamma \rangle.$$

The agent set is $\mathcal{N} = [N]$, the state space is $\mathcal{S}$, and the joint action space factors as

$$\mathcal{A} = \mathcal{A}^1 \times \dots \times \mathcal{A}^N.$$

For agent i , write $\mathcal{A}^{-i} := \prod_{j \neq i} \mathcal{A}^j$. We write a joint action as $a = (a^1, \dots, a^N)$ and as $a = (a^i, a^{-i})$ when focusing on agent i . The transition kernel is $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, the initial distribution is $\mu \in \Delta(\mathcal{S})$, and the discount factor is $\gamma \in [0, 1]$. Each normalized reward function is $r^i : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$.

All policies are stationary product policies $\pi = (\pi^1, \dots, \pi^N)$, where

$$\pi^i : \mathcal{S} \rightarrow \Delta(\mathcal{A}^i), \quad \pi(a | s) = \prod_{i=1}^N \pi^i(a^i | s).$$

For agent i , define

$$\begin{aligned} G_t^i &:= \sum_{\ell=0}^{\infty} \gamma^\ell r^i(s_{t+\ell}, a_{t+\ell}), \\ V_\pi^i(s) &:= \mathbb{E}_\pi[G_t^i | s_t = s], \\ Q_\pi^i(s, a) &:= \mathbb{E}_\pi[G_t^i | s_t = s, a_t = a], \\ A_\pi^i(s, a) &:= Q_\pi^i(s, a) - V_\pi^i(s), \\ V_\pi^i(\mu) &:= \mathbb{E}_{s \sim \mu}[V_\pi^i(s)]. \end{aligned}$$

For a local action $b \in \mathcal{A}^i$, the marginalized local advantage is

$$A_\pi^i(s, b) := \sum_{a^{-i} \in \mathcal{A}^{-i}} \pi^{-i}(a^{-i} | s) A_\pi^i(s, b, a^{-i}). \quad (1)$$

The unnormalized discounted state-visitation measure is

$$d_\mu^\pi(s) := \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s | s_0) \right], \quad \sum_{s \in \mathcal{S}} d_\mu^\pi(s) = \frac{1}{1 - \gamma}.$$

1.2 MARKOV POTENTIAL GAMES

Definition 1 (Markov potential game). A Markov game is a *Markov potential game* (MPG) if there is a function $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ such that, for every agent i , state s , opponents' policy π^{-i} , and pair of

local policies $\pi^i, \tilde{\pi}^i$,

$$\begin{aligned} & V_{\tilde{\pi}^i, \pi^{-i}}^i(s) - V_{\pi^i, \pi^{-i}}^i(s) \\ &= \mathbb{E}_{(\tilde{\pi}^i, \pi^{-i})} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| s_0 = s \right] - \mathbb{E}_{(\pi^i, \pi^{-i})} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| s_0 = s \right]. \end{aligned}$$

For a product policy π , define its total potential by

$$\begin{aligned} \Phi_{\pi}(s) &:= \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| s_0 = s \right], \\ \Phi_{\pi}(\mu) &:= \mathbb{E}_{s \sim \mu} [\Phi_{\pi}(s)]. \end{aligned}$$

The MPG condition is equivalently

$$V_{\tilde{\pi}^i, \pi^{-i}}^i(s) - V_{\pi^i, \pi^{-i}}^i(s) = \Phi_{\tilde{\pi}^i, \pi^{-i}}(s) - \Phi_{\pi^i, \pi^{-i}}(s).$$

A fully cooperative game with $r^i \equiv r$ for all agents is an MPG with $\phi = r$.

1.3 NASH GAP AND REGULARITY

Definition 2 (Nash gap and ϵ -Nash policy). The *Nash gap* of a product policy π is

$$\text{Nash gap}_M(\pi) := \max_{i \in \mathcal{N}} \left(\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) \right),$$

where the inner maximum ranges over all stationary local policies $\tilde{\pi}^i : \mathcal{S} \rightarrow \Delta(\mathcal{A}^i)$. A policy is ϵ *Nash* if $\text{Nash gap}_M(\pi) \leq \epsilon$.

Assumption 3 (Standing regularity).

(i) *Positive visitation*. For every product policy π and state s , $d_{\mu}^{\pi}(s) > 0$. Consequently,

$$\kappa := \max_{\pi, \pi'} \left\| d_{\mu}^{\pi} / d_{\mu}^{\pi'} \right\|_{\infty} = \max_{\pi, \pi'} \max_{s \in \mathcal{S}} \frac{d_{\mu}^{\pi}(s)}{d_{\mu}^{\pi'}(s)} < \infty. \quad (2)$$

(ii) *Bounded total potential*. There are finite constants $\Phi_{\min} \leq \Phi_{\max}$ such that

$$\Phi_{\min} \leq \Phi_{\pi}(s) \leq \Phi_{\max} \quad \text{for every product policy } \pi \text{ and state } s.$$

The stationary product-policy space is a finite product of probability simplexes and is therefore compact. The map $\pi \mapsto d_{\mu}^{\pi}(s)$ is continuous for every state s . Consequently, positive visitation for every policy implies a uniformly positive minimum visitation over policies and states, which justifies the finiteness of κ in equation 2.

2 FINITE-GROUP STATE SYMMETRY

Let G be a finite group with identity e . A left action of G on a finite set $\mathcal{X}$ is a family of bijections $\{T_g : \mathcal{X} \rightarrow \mathcal{X}\}_{g \in G}$ satisfying

$$T_e = \text{id}, \quad T_g \circ T_h = T_{gh}.$$

The orbit and stabilizer of $x \in \mathcal{X}$ are

$$\begin{aligned} [x] &:= \{T_g[x] : g \in G\}, \\ \text{Stab}(x) &:= \{g \in G : T_g[x] = x\}. \end{aligned}$$

The orbits partition $\mathcal{X}$ and form the orbit space $\mathcal{X}/G$. The orbit-stabilizer theorem gives

$$|[x]| = \frac{|G|}{|\text{Stab}(x)|} \leq |G|, \quad |\mathcal{X}/G| \geq \frac{|\mathcal{X}|}{|G|}.$$

Definition 4 (Finite-group state symmetry). Let G act on $\mathcal{S}$ through bijections $\{L_g\}_{g \in G}$. A Markov game M is G -state-symmetric if, for every $g \in G$, $s, s' \in \mathcal{S}$, $a \in \mathcal{A}$, and $i \in \mathcal{N}$,

$$\begin{aligned} P(L_g[s'] \mid L_g[s], a) &= P(s' \mid s, a), \\ r^i(L_g[s], a) &= r^i(s, a). \end{aligned} \quad (3)$$

The action a is unchanged: the group acts only on the state coordinate.

Assumption 5 (Invariant potential and initial distribution). For the G -state-symmetric MPG M , assume

$$\begin{aligned} \phi(L_g[s], a) &= \phi(s, a), \\ \mu(L_g[s]) &= \mu(s) \end{aligned}$$

for every $g \in G$, $s \in \mathcal{S}$, and $a \in \mathcal{A}$.

Potential invariance is automatic in a cooperative game when $\phi = r$. Initial-distribution invariance means that states in the same orbit receive equal initial mass.

Definition 6 (G -invariant policy). A product policy π is G invariant if

$$\pi^i(a^i \mid L_g[s]) = \pi^i(a^i \mid s) \quad \text{for every } i, g, s, \text{ and } a^i.$$

Equivalently, $\pi(a \mid L_g[s]) = \pi(a \mid s)$ for every joint action a .

The per-agent conditions imply joint invariance by multiplying the local action probabilities. Conversely, joint invariance implies each per-agent condition after summing over the other agents' actions. Thus an invariant policy assigns the same local action distribution to every state in an orbit. We use the same terminology for a local policy: ν^i is G invariant if $\nu^i(a^i \mid L_g[s]) = \nu^i(a^i \mid s)$ for every g, s, a^i .

3 THE QUOTIENT MARKOV GAME

Let

$$\bar{\mathcal{S}} := \mathcal{S}/G$$

be the set of state orbits, and let

$$\rho : \mathcal{S} \rightarrow \bar{\mathcal{S}}, \quad \rho(s) := [s],$$

be the orbit map. Fix a representative $\iota(\bar{s}) \in \bar{s}$ for each $\bar{s} \in \bar{\mathcal{S}}$.

Definition 7 (Quotient Markov game induced by G). Let M be a G -state-symmetric Markov game. Its *quotient Markov game* induced by G is

$$\bar{M} := \langle \mathcal{N}, \bar{\mathcal{S}}, \mathcal{A}, \bar{P}, \bar{r}, \bar{\mu}, \gamma \rangle.$$

It has the same agents, local action spaces, and discount factor as M . For any $s \in \bar{s}$, its transition kernel, rewards, and initial distribution are

$$\bar{P}(\bar{s}' \mid \bar{s}, a) := \sum_{s' \in \bar{s}'} P(s' \mid s, a), \quad (4)$$

$$\bar{r}^i(\bar{s}, a) := r^i(s, a), \quad (5)$$

$$\bar{\mu}(\bar{s}) := \sum_{s \in \bar{s}} \mu(s).$$

When M is an MPG satisfying Theorem 5, the corresponding orbit-level potential signal is

$$\bar{\phi}(\bar{s}, a) := \phi(s, a) \quad \text{for any } s \in \bar{s}. \quad (6)$$

Quotient-process notation. For a quotient policy $\bar{\pi}$, write $\bar{V}_{\bar{\pi}}^i$ and $\bar{Q}_{\bar{\pi}}^i$ for agent i 's value and action-value functions in $\bar{M}$, and write $\bar{d}_{\bar{\mu}}^{\bar{\pi}}$ for its discounted state-visitation measure from $\bar{\mu}$. These quantities use the same definitions as in Section 1, with $(M, \mathcal{S}, P, r, \mu)$ replaced by $(\bar{M}, \bar{\mathcal{S}}, \bar{P}, \bar{r}, \bar{\mu})$. Let $\Pi(\bar{M})$ denote the stationary product policies on the quotient and let $\Pi_G(M)$ denote the G -invariant stationary product policies on M .

Lifting quotient policies. Let $\bar{\pi}$ be a stationary product policy on $\bar{M}$. Its lift to M is the product policy $\bar{\pi}^\uparrow$ defined coordinatewise by

$$\bar{\pi}^{\uparrow,i}(a^i | s) := \bar{\pi}^i(a^i | \rho(s)).$$

The lift is G invariant because $\rho(L_g[s]) = \rho(s)$.

Conversely, a G -invariant policy π on M can be projected to $\bar{M}$ by

$$\pi^{\downarrow,i}(a^i | \bar{s}) := \pi^i(a^i | \iota(\bar{s})).$$

Invariance makes this definition independent of the selected representative. Thus quotient policies are in bijection with G -invariant policies on M .

Black-box rate convention. All rate statements below use a target accuracy $\epsilon > 0$. Fix a learning algorithm. A function $T_{\text{MG}}(\epsilon, S; \vartheta)$ is a *uniform MPG certificate* on a class of games if, for every game X in the class,

$$T \geq T_{\text{MG}}(\epsilon, |S_X|; \vartheta) \implies \min_{0 \leq t < T} \text{Nash gap}_X(\pi_t^X) \leq \epsilon. \quad (7)$$

Here $S = |S_X|$ is the number of Markov states and ϑ collects every other hypothesis and numerical bound used by the convergence theorem. Calling the certificate uniform means that the same ϑ is valid for all games being compared.

Theorem 8 (Exact quotient reduction and unrestricted Nash preservation). *Let M be a G -state-symmetric MPG under Theorems 3 and 5. Then the quotient $\bar{M}$ has the following properties.*

1. *The quotient data in Theorem 7 do not depend on the representative $s \in \bar{s}$. The resulting quotient is an MPG with potential $\bar{\phi}$ and total-potential bounds $[\Phi_{\min}, \Phi_{\max}]$. For every quotient policy $\bar{\pi}$, its lift is G invariant and preserves statewise values:*

$$V_{\bar{\pi}^\uparrow}^i(s) = \bar{V}_{\bar{\pi}}^i(\rho(s)).$$

2. *Lifting preserves the unrestricted Nash gap:*

$$\text{Nash gap}_M(\bar{\pi}^\uparrow) = \text{Nash gap}_{\bar{M}}(\bar{\pi}). \quad (8)$$

The maximization on the left still ranges over all local policies on M . Consequently, $\bar{\pi}$ is ϵ Nash in $\bar{M}$ if and only if $\bar{\pi}^\uparrow$ is ϵ Nash in M .

3. *Suppose the uniform certificate above applies to a class containing both M and $\bar{M}$. Run the algorithm on $\bar{M}$ and lift every iterate as $\pi_t := \bar{\pi}_t^\uparrow$. Then*

$$\min_{0 \leq t < T} \text{Nash gap}_M(\pi_t) \leq \epsilon \quad \text{whenever} \quad T \geq T_{\text{MG}}(\epsilon, |\bar{S}|; \vartheta). \quad (9)$$

Thus any uniform rate for the unreduced MPG transfers by substituting $|\bar{S}|$ for $|S|$. If the certificate is nondecreasing in S , this sufficient iteration budget is no larger than the unreduced one. In particular, if a sufficient budget has state dependence

$$B_X(\epsilon) = C(\vartheta) |S_X|^\alpha \epsilon^{-p}, \quad C(\vartheta) > 0, \quad \alpha, p \geq 0,$$

then the full and quotient sufficient budgets satisfy

$$\frac{B_{\text{quot}}(\epsilon)}{B_{\text{full}}(\epsilon)} = \left(\frac{|\bar{S}|}{|S|} \right)^\alpha. \quad (10)$$

This is a comparison of sufficient upper bounds, not realized runtime.

Proof idea. Transition equivariance makes the orbit process Markov, and a Bellman fixed-point argument identifies quotient values with lifted values. Against invariant opponents, an agent's best-response MDP is itself G symmetric. Its optimal action values are constant on state orbits, so fixed tie-breaking gives an invariant optimal response. Quotient deviations therefore attain the same value as unrestricted deviations in the original game. The rate statement then applies the uniform certificate to the quotient iterates and lifts them. Formal proofs are in Sections A to E.

Remark 9 (Scope of the quotient). The construction uses the same joint action a at every state in an orbit. If the group also transforms action labels or permutes agent identities, then equation 4 does not by itself define the appropriate reduced decision problem. Such symmetries require a reduction that also tracks the moved action or agent coordinates. Homogeneous agent relabelings are treated through pointed-state policy sharing in Section 4.

4 HOMOGENEOUS AGENT-PERMUTATION SYMMETRY

State quotients identify states while keeping every decision coordinate fixed. Relabeling homogeneous agents instead moves state, action, reward, and policy coordinates together. The appropriate exact representation is therefore a shared policy on orbits of *pointed states*, where the point records the acting agent. Let S_N denote the symmetric group of permutations of $\mathcal{N} = [N]$. As in Section 1, each local policy is indexed by the full state; the pointed orbit below is its invariant summary relative to a focal agent.

Assumption 10 (Homogeneous agent-permutation structure). Let $\mathcal{C}$ and $\mathcal{B}$ be finite sets, let $\mathcal{S} = \mathcal{C}^N$, and let $\mathcal{A}^i = \mathcal{B}$ for every agent. For $\sigma \in S_N$, define

$$(\sigma \cdot s)^i := s^{\sigma^{-1}(i)}, \quad (\sigma \cdot a)^i := a^{\sigma^{-1}(i)}.$$

Assume that, for every $\sigma \in S_N$, $i \in \mathcal{N}$, $s, s' \in \mathcal{S}$, and $a \in \mathcal{B}^N$,

$$P(\sigma \cdot s' \mid \sigma \cdot s, \sigma \cdot a) = P(s' \mid s, a), \quad (11)$$

$$r^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = r^i(s, a). \quad (12)$$

For the MPG potential and initial distribution, assume

$$\begin{aligned} \phi(\sigma \cdot s, \sigma \cdot a) &= \phi(s, a), \\ \mu(\sigma \cdot s) &= \mu(s). \end{aligned} \quad (13)$$

Let S_N act on a pointed state by $\sigma \cdot (i, s) := (\sigma(i), \sigma \cdot s)$, and define

$$\mathcal{Z} := (\mathcal{N} \times \mathcal{S})/S_N, \quad z_i(s) := [(i, s)]_{S_N}.$$

Thus $z_i(s)$ is the orbit of the state as seen from focal agent i . In the Cartesian model $\mathcal{S} = \mathcal{C}^N$, this orbit has the concrete representation

$$z_i(s) \longleftrightarrow (s^i, \{\{s^k : k \neq i\}\}).$$

Equivalently, $\mathcal{Z} \cong \mathcal{C} \times \mathcal{M}_{N-1}(\mathcal{C})$, where $\mathcal{M}_m(\mathcal{C})$ denotes the multisets of size m drawn from $\mathcal{C}$. Hence

$$|\mathcal{Z}| = |\mathcal{C}| \binom{|\mathcal{C}| + N - 2}{N - 1}.$$

Thus sharing uses $|\mathcal{Z}|$ local policy blocks instead of the $N |\mathcal{C}|^N$ blocks in an unrestricted product-policy table.

Definition 11 (Permutation-invariant and shared policies). A product policy π is *permutation invariant* if

$$\pi^{\sigma(i)}(b \mid \sigma \cdot s) = \pi^i(b \mid s) \quad \text{for every } \sigma \in S_N, i \in \mathcal{N}, s \in \mathcal{S}, \text{ and } b \in \mathcal{B}. \quad (14)$$

A product policy is *pointed-orbit shared* if there is one rule $\psi : \mathcal{Z} \rightarrow \Delta(\mathcal{B})$ such that

$$\pi^i(b \mid s) = \psi(b \mid z_i(s)) \quad \text{for every } i \in \mathcal{N}, s \in \mathcal{S}, \text{ and } b \in \mathcal{B}. \quad (15)$$

Write $\Pi_{S_N}(M)$ for the permutation-invariant product policies.

For any rule $\chi : \mathcal{Z} \rightarrow \Delta(\mathcal{B})$ and focal agent i , write $\chi \circ z_i$ for the local policy

$$(\chi \circ z_i)(b \mid s) := \chi(b \mid z_i(s)).$$

Shared-solver rate convention. Call an MPG X *permutation symmetric* if it has agent set $\mathcal{N} = [N]$, a common action set $\mathcal{B}_X$, a finite state space $\mathcal{S}_X$ equipped with a S_N action, and satisfies the transition, reward, potential, and initial-distribution symmetry identities in Theorem 10. Let S_N act on pointed states by $\sigma \cdot (i, x) := (\sigma(i), \sigma \cdot x)$, and define

$$\begin{aligned} \mathcal{Z}_X &:= (\mathcal{N} \times \mathcal{S}_X)/S_N, & z_i^X(x) &:= [(i, x)]_{S_N}, \\ \text{OrbGap}_X^i(\pi) &:= \max_{\chi : \mathcal{Z}_X \rightarrow \Delta(\mathcal{B}_X)} \left[V_{\chi \circ z_i^X, \pi^{-i}}^i(\mu_X) - V_{\pi}^i(\mu_X) \right]. \end{aligned} \quad (16)$$

Here $(\chi \circ z_i^X)(b \mid x) := \chi(b \mid z_i^X(x))$, and μ_X is the initial distribution of X . A *pointed-orbit shared solver* keeps every iterate in the class $\pi^i(b \mid x) = \psi(b \mid z_i^X(x))$. For a fixed such solver, a function $T_{\text{sh}}(\epsilon, S, D; \vartheta)$ is a *uniform shared certificate* if, for any fixed agent i ,

$$T \geq T_{\text{sh}}(\epsilon, |S_X|, |\mathcal{Z}_X|; \vartheta) \implies \min_{0 \leq t < T} \text{OrbGap}_X^i(\pi_t) \leq \epsilon. \quad (17)$$

Here $S = |S_X|$, $D = |\mathcal{Z}_X|$, and ϑ again collects all common hypotheses and bounds. For the present game, $\mathcal{Z}_M = \mathcal{Z}$ and $z_i^M = z_i$.

Theorem 12 (Exact agent-permutation sharing and unrestricted best responses). *Let M be an MPG satisfying Theorem 10. Then:*

1. *The concrete pair $(s^i, \{\{s^k : k \neq i\}\})$ uniquely represents $z_i(s)$. Permutation-invariant product policies are exactly pointed-orbit shared policies, so equation 15 is a bijection between rules on $\mathcal{Z}$ and policies in $\Pi_{S_N}(M)$.*
2. *For every $\pi \in \Pi_{S_N}(M)$ and any fixed agent i , against the invariant opponents π^{-i} , an unrestricted optimal stationary response can be chosen in shared form:*

$$\max_{\nu^i: S \rightarrow \Delta(\mathcal{B})} V_{\nu^i, \pi^{-i}}^i(\mu) = \max_{\chi: \mathcal{Z} \rightarrow \Delta(\mathcal{B})} V_{\chi \circ z_i, \pi^{-i}}^i(\mu). \quad (18)$$

The best-response gains are equal across agents. Hence, for every i ,

$$\text{Nash gap}_M(\pi) = \max_{\chi: \mathcal{Z} \rightarrow \Delta(\mathcal{B})} \left[V_{\chi \circ z_i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) \right]. \quad (19)$$

This is the ordinary Nash gap from Theorem 2, whose definition allows every stationary local deviation.

3. *If a pointed-orbit shared solver has the uniform certificate above on a class containing M , then its iterates on M satisfy*

$$\min_{0 \leq t < T} \text{Nash gap}_M(\pi_t) \leq \epsilon \quad \text{whenever} \quad T \geq T_{\text{sh}}(\epsilon, |\mathcal{C}|^N, |\mathcal{Z}|; \vartheta). \quad (20)$$

For a bound comparison, additionally suppose that the rate analysis is orbit compatible: an untied counterpart controls the ordinary Nash gap with the same certificate form when $D = N|S_X|$, the shared implementation controls equation 16 with $D = |\mathcal{Z}_X|$, and both have sufficient budgets of the form

$$B_{\text{mode}}(\epsilon) = C_{\text{mode}}(\vartheta) S^\alpha D_{\text{mode}}^\delta \epsilon^{-p}, \quad \alpha, \delta, p \geq 0,$$

where $D_{\text{full}} = N|S_X|$ and $D_{\text{perm}} = |\mathcal{Z}_X|$. Assume the positive prefactors satisfy $C_{\text{perm}}(\vartheta) \leq C_{\text{full}}(\vartheta)$. Then

$$\frac{B_{\text{perm}}(\epsilon)}{B_{\text{full}}(\epsilon)} \leq \left(\frac{|\mathcal{Z}|}{N|\mathcal{C}|^N} \right)^\delta. \quad (21)$$

Exact symmetry gives the transfer in equation 20; it does not by itself imply orbit compatibility. In particular, constants affected by orbit multiplicities must be absorbed into C_{mode} , and the prefactor inequality is an additional assumption. The ratio compares sufficient iteration bounds, not realized runtime.

Proof idea. A permutation-invariant local policy is constant on the orbit of the pointed state (i, s) , which gives the shared-rule representation. For a fixed agent i , the best-response MDP is symmetric under the permutations that fix i ; their state orbits are exactly the fibers of z_i . A symmetric optimal response therefore exists, while relabeling agents shows that every agent has the same best-response gain. This converts the shared solver’s pointed-orbit gap into the ordinary Nash gap. Formal proofs and supporting equivariance identities are in Sections F to I.

Remark 13 (Sharing rather than a state quotient). The orbit $z_i(s)$ depends on a focal agent. Thus Theorem 12 does not replace the original game by a Markov game on $\mathcal{Z}$ and does not require a unilateral deviator to preserve global permutation invariance. It says that invariant product policies have one shared representation and that, at such a policy, a best response for agent i can be chosen invariant under permutations that fix i .

Remark 14 (Automatic potential invariance). In a fully cooperative game with $r^i \equiv r$, the choice $\phi = r$ is permutation invariant whenever the common reward is. More generally, if $\phi(s, a) = N^{-1} \sum_i r^i(s, a)$ is a valid MPG potential, then equation 12 makes it permutation invariant by reindexing the sum over agents.

5 COMBINING STATE QUOTIENTS AND AGENT-PERMUTATION SHARING

The two reductions can be composed when the state-only symmetry is compatible with agent relabeling. Commutativity is the condition that makes agent permutations well defined after passing to state orbits.

Assumption 15 (Commuting state and agent symmetries). Suppose M satisfies both the G -state-symmetry conditions in Theorems 4 and 5 and the homogeneous agent-permutation conditions in Theorem 10. Assume also that

$$L_g[\sigma \cdot s] = \sigma \cdot L_g[s] \quad \text{for every } g \in G, \sigma \in S_N, \text{ and } s \in \mathcal{S}. \quad (22)$$

Under equation 22, $G \times S_N$ acts on pointed states by

$$(g, \sigma) \cdot (i, s) := (\sigma(i), L_g[\sigma \cdot s]).$$

Define the combined pointed-orbit space and map by

$$\mathcal{W} := (\mathcal{N} \times \mathcal{S}) / (G \times S_N), \quad w_i(s) := [(i, s)]_{G \times S_N}. \quad (23)$$

For $\chi : \mathcal{W} \rightarrow \Delta(\mathcal{B})$, write $(\chi \circ w_i)(b \mid s) := \chi(b \mid w_i(s))$. A product policy is *jointly invariant* if it is both G invariant in the sense of Theorem 6 and permutation invariant in the sense of Theorem 11.

Theorem 16 (Commuting reductions and double transfer). *Let M be an MPG under Theorems 3 and 15. Then:*

1. *Agent permutations descend to the state quotient through*

$$\sigma \cdot \bar{s} := \rho(\sigma \cdot s), \quad s \in \bar{s},$$

and $\bar{M}$ is permutation symmetric under this action. Moreover,

$$\mathcal{W} \cong (\mathcal{N} \times (\mathcal{S}/G)) / S_N \cong \mathcal{Z}/G, \quad (24)$$

where the induced action on $\mathcal{Z}$ is $g \cdot [(i, s)]_{S_N} := [(i, L_g[s])]_{S_N}$.

2. *Jointly invariant policies are exactly the policies represented by one rule $\psi : \mathcal{W} \rightarrow \Delta(\mathcal{B})$ through $\pi^i(b \mid s) = \psi(b \mid w_i(s))$. At every such policy and for every fixed agent i ,*

$$\text{Nash gap}_M(\pi) = \max_{\chi : \mathcal{W} \rightarrow \Delta(\mathcal{B})} \left[V_{\chi \circ w_i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) \right]. \quad (25)$$

Thus the maximization still represents the unrestricted Nash gap.

3. *Let a pointed-orbit shared solver have the uniform certificate $T_{\text{sh}}(\epsilon, S, D; \vartheta)$ from equation 17 on a class containing $\bar{M}$. Run it on $\bar{M}$, whose pointed-orbit space is $\mathcal{W}$ by equation 24, and lift its iterates to M . Then*

$$\min_{0 \leq t < T} \text{Nash gap}_M(\pi_t) \leq \epsilon \quad \text{whenever} \quad T \geq T_{\text{sh}}(\epsilon, |\bar{\mathcal{S}}|, |\mathcal{W}|; \vartheta). \quad (26)$$

Suppose additionally that the orbit-compatible untied/shared certificate from Theorem 12 applies to both M and $\bar{M}$, with common exponents $\alpha, \delta, p \geq 0$. Write the four sufficient budgets as

$$B_{\text{mode}}(\epsilon) = C_{\text{mode}}(\vartheta) S_{\text{mode}}^{\alpha} D_{\text{mode}}^{\delta} \epsilon^{-p},$$

where the state and policy-block counts are

mode	full	perm	quot	both
S_{mode}	$ \mathcal{S} $	$ \mathcal{S} $	$ \bar{\mathcal{S}} $	$ \bar{\mathcal{S}} $
D_{mode}	$N \mathcal{S} $	$ \mathcal{Z} $	$N \bar{\mathcal{S}} $	$ \mathcal{W} $

Here “perm” means permutation sharing only, “quot” means the untied state quotient only, and “both” means their composition. Assume the positive prefactors satisfy

$$C_{\text{both}}(\vartheta) \leq \min \{C_{\text{full}}(\vartheta), C_{\text{perm}}(\vartheta), C_{\text{quot}}(\vartheta)\}.$$

The sufficient bounds then satisfy

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{full}}(\epsilon)} \leq \left(\frac{|\bar{\mathcal{S}}|}{|\mathcal{S}|} \right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|} \right)^\delta \left(\frac{|\mathcal{Z}|}{N|\mathcal{S}|} \right)^\delta, \quad (27)$$

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{perm}}(\epsilon)} \leq \left(\frac{|\bar{\mathcal{S}}|}{|\mathcal{S}|} \right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|} \right)^\delta \leq 1. \quad (28)$$

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{quot}}(\epsilon)} \leq \left(\frac{|\mathcal{W}|}{N|\bar{\mathcal{S}}|} \right)^\delta \leq 1. \quad (29)$$

Thus the first bound separates the permutation-only block reduction from the two additional reductions obtained by quotienting the Markov states and the shared policy blocks; the latter two bounds show that adding either reduction to the other is non-worsening under the stated certificate and prefactor assumptions. The required prefactor inequalities are assumptions on the convergence analysis, not consequences of symmetry. If G acts freely on $\mathcal{S}$ and its induced action on $\mathcal{Z}$ is also free, both $|\bar{\mathcal{S}}|/|\mathcal{S}|$ and $|\mathcal{W}|/|\mathcal{Z}|$ equal $|G|^{-1}$, so

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{perm}}(\epsilon)} \leq |G|^{-(\alpha+\delta)}, \quad (30)$$

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{full}}(\epsilon)} \leq |G|^{-(\alpha+\delta)} \left(\frac{|\mathcal{Z}|}{N|\mathcal{S}|} \right)^\delta. \quad (31)$$

Proof idea. Commutativity lets agent permutations act on G -orbits, so quotienting states and then taking pointed-state orbits identifies the same classes as the direct $G \times S_N$ action. At a jointly invariant policy, a focal agent’s best-response MDP is symmetric under G and the agent permutations that fix that agent. The resulting invariant optimal response is constant on the fibers of w_i . Running the shared solver on the quotient and lifting its iterates then combines the two exact Nash-gap identities. The detailed orbit and rate calculations are given in Section J.

6 GENERATIVE-MODEL SAMPLE COMPLEXITY THROUGH THE REDUCTIONS

The rate transfers in Theorems 8, 12 and 16 concern iteration complexity: they substitute quotient sizes into the certificate of an abstract convergence theorem while the game itself is treated as known. This section gives a second, statistical application of the exact reductions. When the transition kernel is unknown and must be estimated from samples, both symmetries compress the estimation problem, in different factors: the state quotient shrinks the estimand itself, so the sufficient data budget scales with $|\bar{\mathcal{S}}|$ rather than $|\mathcal{S}|$, while agent-permutation symmetry makes entire rows of the kernel relabelings of one another, so the number of rows that must be estimated drops from exponential to polynomial in N . The two improvements compound under Theorem 15.

Sampling model. Fix a G -state-symmetric MPG M satisfying Theorems 3 and 5. The learner has *generative-model* access to M : for any queried pair $(s, a) \in \mathcal{S} \times \mathcal{A}$, it observes an independent draw $s' \sim P(\cdot | s, a)$. The learner knows the group action, hence the orbit map ρ and the representatives ι , and can evaluate the potential ϕ at queried pairs; the kernel P is unknown. In the fully cooperative case $r^i \equiv r$ with $\phi = r$, evaluating ϕ amounts to observing the reward, so the last requirement is automatic; Theorem 24 treats estimated potentials. This access model is *centralized*, as is standard in model-based sample-complexity analyses of Markov games (Zhang et al., 2020; Subramanian et al., 2023): a single learner selects joint queries, evaluates the global potential, plans over joint actions, and dispatches the resulting joint policy. Theorem 23 discusses decentralization.

Pooled orbit-level estimation. For each quotient pair $(\bar{s}, a) \in \bar{\mathcal{S}} \times \mathcal{A}$, query the representative pair $(\iota(\bar{s}), a)$ exactly n times, observe independent draws $s'_1, \dots, s'_n \sim P(\cdot | \iota(\bar{s}), a)$, and set

$$\hat{P}(\bar{s}' | \bar{s}, a) := \frac{1}{n} \sum_{k=1}^n \mathbf{1}\{s'_k \in \bar{s}'\}. \quad (32)$$

Because $\bar{P}(\bar{s}' \mid \bar{s}, a) = \Pr[s' \in \bar{s}']$ for a draw $s' \sim P(\cdot \mid \iota(\bar{s}), a)$, the estimator is unbiased for the quotient kernel, and by Theorem 25 the queried representative may be replaced by any orbit member without changing the estimand. The total query budget is $n|\bar{\mathcal{S}}||\mathcal{A}|$.

Certainty-equivalent potential planning. The *potential MDP* of the quotient is the single-controller MDP with state space $\bar{\mathcal{S}}$, action space $\mathcal{A}$, reward $\bar{\phi}$, kernel $\bar{P}$, and discount γ ; the value function it assigns to a stationary product policy $\bar{\pi}$ is the total potential $\bar{\Phi}_{\bar{\pi}}$. The *empirical potential MDP* replaces $\bar{P}$ by $\hat{P}$, and $\hat{\Phi}_{\bar{\pi}}$ denotes its value functions. The planner computes a stationary deterministic policy $\hat{\pi}_{\text{ce}} : \bar{\mathcal{S}} \rightarrow \mathcal{A}$ that is optimal for the empirical potential MDP, for instance by value iteration on $\hat{P}$ followed by greedy selection. A stationary deterministic joint policy assigns one joint action to each state and is therefore a product policy, so $\bar{\pi}_{\text{ce}} \in \Pi(M)$ and its lift $\bar{\pi}_{\text{ce}}^\uparrow$ is a G -invariant product policy on M . Recall $\Phi_{\min}, \Phi_{\max}$ from Theorem 3 and set

$$\Phi_{\text{sup}} := \max\{|\Phi_{\min}|, |\Phi_{\max}|\}.$$

Theorem 17 (Generative-model sample complexity through the quotient). *Let M be a G -state-symmetric MPG under Theorems 3 and 5, let $\epsilon > 0$ and $\delta \in (0, 1)$, and let $\bar{\pi}_{\text{ce}}$ be computed from equation 32 as above with per-pair budget n .*

1. If

$$n \geq \frac{8\gamma^2\Phi_{\text{sup}}^2}{(1-\gamma)^2\epsilon^2} \left(|\bar{\mathcal{S}}| \log 2 + \log \frac{|\bar{\mathcal{S}}||\mathcal{A}|}{\delta} \right), \quad (33)$$

then, with probability at least $1 - \delta$ over the draws in equation 32,

$$\text{Nash gap}_M(\bar{\pi}_{\text{ce}}^\uparrow) \leq \epsilon. \quad (34)$$

Consequently, the total number of generative-model queries that suffices is

$$B^{\text{gen}}(\epsilon, \delta; S, A) := \frac{8\gamma^2\Phi_{\text{sup}}^2}{(1-\gamma)^2\epsilon^2} SA \left(S \log 2 + \log \frac{SA}{\delta} \right), \quad (35)$$

evaluated at $S = |\bar{\mathcal{S}}|$ and $A = |\mathcal{A}|$.

2. The same construction applied with the trivial group $G = \{e\}$ runs directly on M , uses no symmetry, and attains equation 34 with sufficient budget $B^{\text{gen}}(\epsilon, \delta; |\mathcal{S}|, |\mathcal{A}|)$. The two sufficient budgets satisfy

$$\frac{B^{\text{gen}}(\epsilon, \delta; |\bar{\mathcal{S}}|, |\mathcal{A}|)}{B^{\text{gen}}(\epsilon, \delta; |\mathcal{S}|, |\mathcal{A}|)} = \frac{|\bar{\mathcal{S}}|}{|\mathcal{S}|} \cdot \frac{|\bar{\mathcal{S}}| \log 2 + \log(|\bar{\mathcal{S}}||\mathcal{A}|/\delta)}{|\mathcal{S}| \log 2 + \log(|\mathcal{S}||\mathcal{A}|/\delta)} \leq \frac{|\bar{\mathcal{S}}|}{|\mathcal{S}|}. \quad (36)$$

When the state-count terms dominate the logarithmic terms in both parentheses, the ratio is $(|\bar{\mathcal{S}}|/|\mathcal{S}|)^2$ up to that logarithmic correction; if G acts freely on $\mathcal{S}$, then $|\bar{\mathcal{S}}|/|\mathcal{S}| = |G|^{-1}$. As elsewhere, this compares sufficient sample-size bounds, not realized sample usage.

Proof idea. Pooling makes each quotient row estimable at a single representative, and an ℓ_1 concentration bound holds uniformly over quotient pairs. A simulation lemma for the potential MDP converts kernel error into a uniform error on total potentials, so the certainty-equivalent policy nearly maximizes the true quotient potential statewise. In an MPG, every unilateral deviation gain is a potential difference and is therefore controlled by the potential suboptimality, which bounds the quotient Nash gap. Exact Nash-gap preservation in Theorem 8 transfers the guarantee to M unchanged. The full argument is in Section K.

Pooling over state-action orbits. The quotient construction shrinks the game being estimated. Agent-permutation symmetry admits no reduced game (Theorem 13), yet it still compresses the estimation problem: entire rows of the kernel are relabelings of one another. Let X be a permutation-symmetric MPG in the sense of the shared-solver rate convention of Section 4, satisfying Theorem 3; the leading case is $X = M$ under Theorem 10. Let S_N act diagonally on state-action pairs by $\sigma \cdot (x, a) := (\sigma \cdot x, \sigma \cdot a)$, and define the pair-orbit space and orbit map

$$\mathcal{V}_X := (\mathcal{S}_X \times \mathcal{B}_X^N)/S_N, \quad y(x, a) := [(x, a)]_{S_N}.$$

In the Cartesian model of Theorem 10, a pair orbit is the multiset $\{(s^k, a^k) : k \in [N]\}$, so $\mathcal{Y}_M \cong \mathcal{M}_N(\mathcal{C} \times \mathcal{B})$ and

$$|\mathcal{Y}_M| = \binom{|\mathcal{C}| |\mathcal{B}| + N - 1}{N},$$

polynomial in N of degree $|\mathcal{C}| |\mathcal{B}| - 1$, versus the $|\mathcal{S}| |\mathcal{A}| = (|\mathcal{C}| |\mathcal{B}|)^N$ untied rows.

Transported estimation. Fix a representative $j(\bar{y}) \in \bar{y}$ for each $\bar{y} \in \mathcal{Y}_X$ and, for every pair (x, a) , a transporting permutation $\tau_{x,a} \in S_N$ with $\tau_{x,a} \cdot j(y(x, a)) = (x, a)$, choosing $\tau_{x,a} = e$ whenever (x, a) is itself the representative. For each orbit $\bar{y}$, query its representative pair n times, observe independent draws $x'_1, \dots, x'_n \sim P_X(\cdot | j(\bar{y}))$, and let $\hat{P}_X(\cdot | j(\bar{y}))$ be the resulting empirical row. The *transported estimator* extends the representative rows to every pair:

$$\tilde{P}(x' | x, a) := \hat{P}_X(\tau_{x,a}^{-1} \cdot x' | j(y(x, a))). \quad (37)$$

The total query budget is $n |\mathcal{Y}_X|$. Potential invariance likewise determines ϕ_X everywhere from its values at the $|\mathcal{Y}_X|$ representatives. The planner computes a stationary deterministic policy π_{ce} optimal for the empirical potential MDP of X with kernel $\tilde{P}$; as before, a stationary deterministic joint policy is a product policy.

Theorem 18 (Generative-model sample complexity through permutation pooling). *Let X be a permutation-symmetric MPG satisfying Theorem 3, let $\epsilon > 0$ and $\delta \in (0, 1)$, and let π_{ce} be computed from equation 37 with per-orbit budget n .*

1. If

$$n \geq \frac{8\gamma^2 \Phi_{\sup}^2}{(1-\gamma)^2 \epsilon^2} \left(|\mathcal{S}_X| \log 2 + \log \frac{|\mathcal{Y}_X|}{\delta} \right), \quad (38)$$

then, with probability at least $1 - \delta$, $\text{Nash gap}_X(\pi_{ce}) \leq \epsilon$. The sufficient total query budget is

$$B^{\text{pool}}(\epsilon, \delta; X) := \frac{8\gamma^2 \Phi_{\sup}^2}{(1-\gamma)^2 \epsilon^2} |\mathcal{Y}_X| \left(|\mathcal{S}_X| \log 2 + \log \frac{|\mathcal{Y}_X|}{\delta} \right). \quad (39)$$

2. The untied construction of Theorem 172. applied to X has sufficient budget $B^{\text{gen}}(\epsilon, \delta; |\mathcal{S}_X|, |\mathcal{B}_X|^N)$. The sufficient budgets satisfy

$$\frac{B^{\text{pool}}(\epsilon, \delta; X)}{B^{\text{gen}}(\epsilon, \delta; |\mathcal{S}_X|, |\mathcal{B}_X|^N)} = \frac{|\mathcal{Y}_X|}{|\mathcal{S}_X| |\mathcal{B}_X|^N} \cdot \frac{|\mathcal{S}_X| \log 2 + \log(|\mathcal{Y}_X|/\delta)}{|\mathcal{S}_X| \log 2 + \log(|\mathcal{S}_X| |\mathcal{B}_X|^N/\delta)} \leq \frac{|\mathcal{Y}_X|}{|\mathcal{S}_X| |\mathcal{B}_X|^N}. \quad (40)$$

Unlike equation 36, the leading terms $|\mathcal{S}_X| \log 2$ coincide because the next-state space does not shrink, so the ratio equals the row-count ratio up to a correction confined to the lower-order logarithm. For $X = M$ in the Cartesian model, the row-count ratio is $\binom{|\mathcal{C}| |\mathcal{B}| + N - 1}{N} / (|\mathcal{C}| |\mathcal{B}|)^N$. As elsewhere, this compares sufficient sample-size bounds, not realized sample usage.

Corollary 19 (Combined pooling through the quotient). *Let M satisfy Theorems 3 and 15. Then, with probability at least $1 - \delta$,*

$$\text{Nash gap}_M(\bar{\pi}_{ce}^\uparrow) \leq \epsilon \quad \text{whenever} \quad n \geq \frac{8\gamma^2 \Phi_{\sup}^2}{(1-\gamma)^2 \epsilon^2} \left(|\bar{\mathcal{S}}| \log 2 + \log \frac{|\mathcal{Y}_{\bar{M}}|}{\delta} \right),$$

where now $\bar{\pi}_{ce}$ is the certainty-equivalent policy computed from the transported estimator on $X = \bar{M}$, whose generative model is obtained from that of M by querying orbit representatives and recording next-state orbits as in equation 32, and the total query budget is $n |\mathcal{Y}_{\bar{M}}|$. The two reductions compound in different factors: the quotient shrinks the next-state space from $|\mathcal{S}|$ to $|\bar{\mathcal{S}}|$, while pooling shrinks the row count from $|\bar{\mathcal{S}}| |\mathcal{A}|$ to $|\mathcal{Y}_{\bar{M}}|$.

Proof idea. Fixing one transporting permutation per pair turns the symmetry identity equation 11 into an exact reconstruction rule: transported rows have the same ℓ_1 error as their representative rows, so uniform concentration is needed only over the $|\mathcal{Y}_X|$ representatives. The simulation, optimality-transfer, and Nash-gap lemmas of the quotient argument apply verbatim because they use only a uniform kernel error and the total-potential bounds; the corollary applies the pooled pipeline to $\bar{M}$, which is permutation symmetric by Theorem 16, and adds the exact lift of Theorem 8. Details are in Section K.

Remark 20 (The estimated models need not be MPGs). Sampling noise breaks the exact potential structure: the games obtained by equipping $\bar{M}$ with $\hat{P}$, or X with $\tilde{P}$, generally admit no potential, so a solver whose certificate presupposes membership in the MPG class cannot simply be run on the estimated game. Certainty-equivalent potential planning sidesteps this: value iteration on the empirical potential MDP is a single-agent computation that does not require the estimated game to be a potential game, and the analysis compares total potentials only under the true kernel. For the transported estimator this also means that exact symmetry of $\tilde{P}$ is unnecessary: when a pair has a nontrivial stabilizer, the empirical representative row need not be stabilizer invariant, so $\tilde{P}$ is determined by the chosen transports rather than being exactly S_N invariant, and the argument never uses invariance of the estimate. Alternatively, a model-to-Nash-gap stability lemma would let an arbitrary MPG solver run on a nearby exact model; the present constructions avoid the need for such a lemma.

Remark 21 (Trajectory data and orbit-level coverage). With trajectory rather than generative access, the estimator equation 32 extends by pooling every observed transition from any state in the orbit $\bar{s}$ under action a ; representative independence (Theorem 25) again makes the pooled estimand equal to $\bar{P}(\cdot | \bar{s}, a)$. The pair $(\bar{s}, a)$ is then estimable as soon as the data visit any orbit member, so coverage requirements weaken from state-action coverage to orbit-action coverage. Quantifying this under a specific data-collection process requires a concentration argument for dependent visits and is not pursued here.

Remark 22 (Scope: statistical versus computational savings). The savings established in this section are statistical. In every mode, the planning step enumerates joint actions, so its computational cost scales with the action count, which equals $|\mathcal{B}|^N$ under Theorem 10. Moreover, the policy returned by transported-estimation planning is a deterministic product policy that need not be permutation invariant: a deterministic invariant policy can be strictly suboptimal for the potential MDP when agents share identical local states, so requiring shared output may conflict with deterministic planning. This is consistent with Theorem 13: permutation symmetry is exploited here for estimation, not through a reduced game. A planner that exploits action symmetry, and shared or mixed certainty-equivalent outputs, are left open.

Remark 23 (Toward a decentralized generative model). Every ingredient of the pipeline is centralized: query selection is joint, the potential is a global quantity that no single agent observes outside the cooperative case $\phi = r$, planning is a single computation over joint actions, and executing the returned deterministic policy requires assigning roles. This matches the system-planner convention of model-based sample-complexity analyses for Markov games (Zhang et al., 2020; Subramanian et al., 2023). The variant in which each agent draws its own samples and computes its own solution is identified as open by Subramanian et al. (2023), the obstruction being that independently estimated games can have different equilibria. Existing decentralized guarantees for MPGs instead assume on-line trajectory access (Leonardos et al., 2022; Fox et al., 2022; Ding et al., 2022; Zhang et al., 2021; Maheshwari et al., 2022), extended recently to partial observability (Jordan et al., 2026), or target weaker equilibrium notions (Jin et al., 2021; Mao et al., 2022), or require coordination between players (Song et al., 2022). The present pipeline suggests an intermediate regime: certainty-equivalent potential planning with a fixed tie-breaking rule is a deterministic map from data to joint policy, so under broadcast samples or shared randomness every agent computes the same joint policy and executes its own coordinate, which removes the equilibrium-selection ambiguity while decentralizing execution. Fully independent sampling remains open, and Theorem 22 adds a symmetric obstruction: a shared deterministic rule can be strictly suboptimal, so a fully decentralized symmetric planner would need mixed shared rules or a correlation device.

Remark 24 (Estimated potentials). If ϕ is bounded and only sample access to it is available, replacing $\bar{\phi}$ by an empirical potential with uniform error ε_ϕ adds $2\varepsilon_\phi/(1 - \gamma)$ to the guarantee equation 34 through the same simulation argument, a lower-order modification.

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A WELL-POSEDNESS OF THE QUOTIENT MARKOV GAME

Lemma 25 (Well-posed quotient data). *Let M be a G -state-symmetric MPG satisfying Theorem 5. Then the quotient data in Theorem 7 are independent of the representative $s \in \bar{s}$. Moreover, $\bar{P}(\cdot \mid \bar{s}, a) \in \Delta(\bar{\mathcal{S}})$ for every $(\bar{s}, a)$ and $\bar{\mu} \in \Delta(\bar{\mathcal{S}})$.*

Proof. Let $s, \tilde{s} \in \bar{s}$. There is a $g \in G$ such that $\tilde{s} = L_g[s]$. For every next-state orbit $\bar{s}'$, the map $u \mapsto L_g[u]$ is a bijection from $\bar{s}'$ to itself. Reindexing the orbit sum and using transition equivariance,

$$\begin{aligned} \sum_{u \in \bar{s}'} P(u \mid \tilde{s}, a) &= \sum_{u \in \bar{s}'} P(L_g[u] \mid L_g[s], a) \\ &= \sum_{u \in \bar{s}'} P(u \mid s, a). \end{aligned}$$

Thus equation 4 is independent of the representative. Reward and potential invariance similarly give

$$r^i(\tilde{s}, a) = r^i(s, a), \quad \phi(\tilde{s}, a) = \phi(s, a),$$

so equation 5 and equation 6 are also representative-independent. The definition of $\bar{\mu}$ already sums over the full orbit and contains no representative choice.

The state orbits partition $\mathcal{S}$. Hence, for any representative $s \in \bar{s}$,

$$\begin{aligned} \sum_{\bar{s}' \in \bar{\mathcal{S}}} \bar{P}(\bar{s}' \mid \bar{s}, a) &= \sum_{\bar{s}' \in \bar{\mathcal{S}}} \sum_{s' \in \bar{s}'} P(s' \mid s, a) \\ &= \sum_{s' \in \mathcal{S}} P(s' \mid s, a) = 1. \end{aligned}$$

Every term is nonnegative, so $\bar{P}(\cdot \mid \bar{s}, a) \in \Delta(\bar{\mathcal{S}})$. Likewise,

$$\sum_{\bar{s} \in \bar{\mathcal{S}}} \bar{\mu}(\bar{s}) = \sum_{\bar{s} \in \bar{\mathcal{S}}} \sum_{s \in \bar{s}} \mu(s) = \sum_{s \in \mathcal{S}} \mu(s) = 1,$$

and $\bar{\mu}$ is nonnegative. Thus $\bar{\mu} \in \Delta(\bar{\mathcal{S}})$. $\square$

B POLICY CORRESPONDENCE AND VALUE EQUIVALENCE

Lemma 26 (Policy correspondence and value equivalence). *Let M be a G -state-symmetric Markov game and let $\bar{M}$ be its quotient. Then the lift and projection satisfy:*

(a) *The maps $\bar{\pi} \mapsto \bar{\pi}^\uparrow$ and $\pi \mapsto \pi^\downarrow$ are mutually inverse bijections between $\Pi(\bar{M})$ and $\Pi_G(M)$.*

(b) *For every $\bar{\pi} \in \Pi(\bar{M})$, agent i , state s , and joint action a ,*

$$V_{\bar{\pi}^\uparrow}^i(s) = \bar{V}_{\bar{\pi}}^i(\rho(s)), \quad Q_{\bar{\pi}^\uparrow}^i(s, a) = \bar{Q}_{\bar{\pi}}^i(\rho(s), a). \quad (41)$$

Consequently, $V_{\bar{\pi}^\uparrow}^i(\mu) = \bar{V}_{\bar{\pi}}^i(\bar{\mu})$.

(c) *For every $\bar{\pi} \in \Pi(\bar{M})$ and $\bar{s} \in \bar{\mathcal{S}}$,*

$$\bar{d}_{\bar{\mu}}^{\bar{\pi}}(\bar{s}) = \sum_{s \in \bar{s}} d_{\mu}^{\bar{\pi}^\uparrow}(s). \quad (42)$$

Proof. For every $g \in G$, $\rho(L_g[s]) = \rho(s)$. Therefore

$$\bar{\pi}^{\uparrow, i}(a^i \mid L_g[s]) = \bar{\pi}^i(a^i \mid \rho(L_g[s])) = \bar{\pi}^i(a^i \mid \rho(s)) = \bar{\pi}^{\uparrow, i}(a^i \mid s),$$

so every lift is G invariant. The projection is well defined because an invariant policy is constant on each orbit. Direct substitution gives

$$(\bar{\pi}^\uparrow)^\downarrow = \bar{\pi}, \quad (\pi^\downarrow)^\uparrow = \pi \quad \text{for every } \pi \in \Pi_G(M).$$

This proves part (a).

Fix a quotient policy $\bar{\pi}$ and write $\pi = \bar{\pi}^\dagger$. Define

$$\tilde{V}^i(s) := \bar{V}_{\bar{\pi}}^i(\rho(s)).$$

Using $\pi(a | s) = \bar{\pi}(a | \rho(s))$ and the quotient definitions,

$$\begin{aligned} \tilde{V}^i(s) &= \sum_a \bar{\pi}(a | \rho(s)) \left[\bar{r}^i(\rho(s), a) + \gamma \sum_{\bar{s}'} \bar{P}(\bar{s}' | \rho(s), a) \bar{V}_{\bar{\pi}}^i(\bar{s}') \right] \\ &= \sum_a \pi(a | s) \left[r^i(s, a) + \gamma \sum_{\bar{s}'} \sum_{s' \in \bar{s}'} P(s' | s, a) \tilde{V}^i(s') \right] \\ &= \sum_a \pi(a | s) \left[r^i(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, a) \tilde{V}^i(s') \right]. \end{aligned}$$

Thus $\tilde{V}^i$ is a fixed point of the Bellman evaluation operator for π on M . That operator is a γ -contraction, so its fixed point is unique and $\tilde{V}^i = V_{\pi}^i$. Substituting this value identity into the one-step action-value equations gives

$$\begin{aligned} Q_{\pi}^i(s, a) &= r^i(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\pi}^i(s') \\ &= \bar{r}^i(\rho(s), a) + \gamma \sum_{\bar{s}'} \bar{P}(\bar{s}' | \rho(s), a) \bar{V}_{\bar{\pi}}^i(\bar{s}') \\ &= \bar{Q}_{\bar{\pi}}^i(\rho(s), a). \end{aligned}$$

This proves equation 41. Averaging its value identity over $s \sim \mu$ and grouping by state orbit yields

$$\begin{aligned} V_{\bar{\pi}^\dagger}^i(\mu) &= \sum_{\bar{s}} \sum_{s \in \bar{s}} \mu(s) \bar{V}_{\bar{\pi}}^i(\bar{s}) \\ &= \sum_{\bar{s}} \bar{\mu}(\bar{s}) \bar{V}_{\bar{\pi}}^i(\bar{s}) = \bar{V}_{\bar{\pi}}^i(\bar{\mu}). \end{aligned}$$

For part (c), the state-to-state kernels induced by π and $\bar{\pi}$ satisfy

$$\sum_{s' \in \bar{s}'} P^\pi(s' | s) = \bar{P}^{\bar{\pi}}(\bar{s}' | \rho(s)). \quad (43)$$

Indeed, this follows by summing $P^\pi(s' | s) = \sum_a \pi(a | s) P(s' | s, a)$ over $\bar{s}'$ and applying equation 4. Let $p_t^\pi(s' | s_0)$ and $\bar{p}_t^{\bar{\pi}}(\bar{s}' | \bar{s}_0)$ denote the corresponding t -step transition probabilities. We claim that

$$\sum_{s' \in \bar{s}'} p_t^\pi(s' | s_0) = \bar{p}_t^{\bar{\pi}}(\bar{s}' | \rho(s_0)) \quad \text{for every } t \geq 0. \quad (44)$$

At $t = 0$, both sides are the indicator of $\rho(s_0) = \bar{s}'$. If the identity holds at time t , then grouping the one-step recursion for p_{t+1}^π by the orbit $\bar{u}$ of the intermediate state and using equation 43 gives

$$\begin{aligned} \sum_{s' \in \bar{s}'} p_{t+1}^\pi(s' | s_0) &= \sum_{\bar{u} \in \bar{\mathcal{S}}} \left(\sum_{u \in \bar{u}} p_t^\pi(u | s_0) \right) \bar{P}^{\bar{\pi}}(\bar{s}' | \bar{u}) \\ &= \sum_{\bar{u} \in \bar{\mathcal{S}}} \bar{p}_t^{\bar{\pi}}(\bar{u} | \rho(s_0)) \bar{P}^{\bar{\pi}}(\bar{s}' | \bar{u}) \\ &= \bar{p}_{t+1}^{\bar{\pi}}(\bar{s}' | \rho(s_0)). \end{aligned}$$

Thus equation 44 follows by induction. Finally,

$$\begin{aligned}\bar{d}_\mu^\pi(\bar{s}') &= \sum_{t=0}^{\infty} \gamma^t \sum_{\bar{s}_0} \bar{\mu}(\bar{s}_0) \bar{p}_t^\pi(\bar{s}' | \bar{s}_0) \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{s_0 \in \mathcal{S}} \mu(s_0) \sum_{s' \in \bar{s}'} p_t^\pi(s' | s_0) \\ &= \sum_{s' \in \bar{s}'} d_\mu^\pi(s'),\end{aligned}$$

where the second equality uses $\bar{\mu}(\bar{s}_0) = \sum_{s_0 \in \bar{s}_0} \mu(s_0)$ and equation 44. This is equation 42. $\square$

C THE QUOTIENT IS A MARKOV POTENTIAL GAME

Lemma 27 (Inherited MPG structure and regularity). *Let M be a G -state-symmetric MPG satisfying Theorems 3 and 5. Then its quotient has the following properties:*

- (a) *It is an MPG with potential $\bar{\phi}$ and total-potential bounds $[\Phi_{\min}, \Phi_{\max}]$.*
- (b) *Every quotient policy has positive discounted visitation on every quotient state.*
- (c) *Its mismatch coefficient*

$$\kappa_G := \max_{\bar{\pi}, \bar{\pi}' \in \Pi(\bar{M})} \left\| \bar{d}_\mu^\pi / \bar{d}_\mu^{\pi'} \right\|_\infty$$

satisfies $\kappa_G \leq \kappa$.

Proof. Let $\bar{\Phi}_\pi$ denote the discounted total potential generated by $\bar{\phi}$ in $\bar{M}$. The Bellman argument in Theorem 26(b), with ϕ and $\bar{\phi}$ in place of r^i and $\bar{r}^i$, gives

$$\Phi_{\bar{\pi}^\uparrow}(s) = \bar{\Phi}_\pi(\rho(s)). \quad (45)$$

Fix an agent i , a quotient policy $\bar{\pi}$, and an alternative local policy $\bar{\nu}^i$. Set $\tilde{\pi} := (\bar{\nu}^i, \bar{\pi}^{-i})$. Lifting acts coordinatewise, so $\tilde{\pi}^\uparrow$ and $\bar{\pi}^\uparrow$ differ only in agent i 's policy. At any representative $s \in \bar{s}$, value equivalence, the MPG identity of M , and equation 45 give

$$\begin{aligned}\bar{V}_{\tilde{\pi}}^i(\bar{s}) - \bar{V}_{\bar{\pi}}^i(\bar{s}) &= V_{\tilde{\pi}^\uparrow}^i(s) - V_{\bar{\pi}^\uparrow}^i(s) \\ &= \Phi_{\tilde{\pi}^\uparrow}(s) - \Phi_{\bar{\pi}^\uparrow}(s) \\ &= \bar{\Phi}_{\tilde{\pi}}(\bar{s}) - \bar{\Phi}_{\bar{\pi}}(\bar{s}).\end{aligned}$$

Hence $\bar{\phi}$ is a potential for $\bar{M}$. Moreover,

$$\bar{\Phi}_{\bar{\pi}}(\bar{s}) = \Phi_{\bar{\pi}^\uparrow}(s) \in [\Phi_{\min}, \Phi_{\max}],$$

which proves part (a).

By equation 42,

$$\bar{d}_\mu^\pi(\bar{s}) = \sum_{s \in \bar{s}} d_\mu^{\pi^\uparrow}(s).$$

Each summand is positive by Theorem 3, proving part (b).

For quotient policies $\bar{\pi}, \bar{\pi}'$ and a quotient state $\bar{s}$,

$$\begin{aligned}\frac{\bar{d}_\mu^\pi(\bar{s})}{\bar{d}_\mu^{\pi'}(\bar{s})} &= \frac{\sum_{s \in \bar{s}} d_\mu^{\pi^\uparrow}(s)}{\sum_{s \in \bar{s}} d_\mu^{\pi'^\uparrow}(s)} \\ &\leq \max_{s \in \bar{s}} \frac{d_\mu^{\pi^\uparrow}(s)}{d_\mu^{\pi'^\uparrow}(s)} \leq \kappa.\end{aligned}$$

The first inequality is the mediant inequality for positive numbers; the second is the definition of κ . Maximizing over quotient states and policies proves $\kappa_G \leq \kappa$. $\square$

D NASH-GAP PRESERVATION

The key point is that, against invariant opponents, an agent has an invariant optimal best response even though the original Nash gap allows all local policies.

Lemma 28 (Invariant optimal best response). *Let M be a G -state-symmetric Markov game. Fix an agent i and a G -invariant product policy π . The best-response MDP faced by agent i when the other agents play π^{-i} is G -state-symmetric and has a G -invariant optimal stationary policy. Consequently,*

$$\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) = \max_{\tilde{\pi}^i \text{ } G\text{-invariant}} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu). \quad (46)$$

Proof. The best-response MDP has state space $\mathcal{S}$, action space $\mathcal{A}^i$, discount γ , initial distribution μ , and data

$$\begin{aligned} P_{\pi^{-i}}^i(s' | s, a^i) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) P(s' | s, a^i, a^{-i}), \\ r_{\pi^{-i}}^i(s, a^i) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) r^i(s, a^i, a^{-i}). \end{aligned}$$

Since π^{-i} is invariant and the group leaves every action label fixed,

$$\begin{aligned} P_{\pi^{-i}}^i(L_g[s'] | L_g[s], a^i) &= \sum_{a^{-i}} \pi^{-i}(a^{-i} | L_g[s]) P(L_g[s'] | L_g[s], a^i, a^{-i}) \\ &= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) P(s' | s, a^i, a^{-i}) \\ &= P_{\pi^{-i}}^i(s' | s, a^i), \end{aligned}$$

where the second equality uses policy invariance and equation 3. The same calculation with rewards gives

$$r_{\pi^{-i}}^i(L_g[s], a^i) = r_{\pi^{-i}}^i(s, a^i).$$

Thus the best-response MDP is G -state-symmetric.

Let $\mathcal{T}$ be its Bellman optimality operator. If a function V is G invariant, then transition and reward symmetry imply

$$(\mathcal{T}V)(L_g[s]) = (\mathcal{T}V)(s).$$

Starting value iteration from the zero function therefore produces only invariant functions. Because $\mathcal{T}$ is a γ -contraction, these iterates converge to the unique optimal value V^* , which is invariant. Its optimal action-value function consequently satisfies

$$Q^*(L_g[s], a^i) = Q^*(s, a^i).$$

Choose a greedy deterministic policy and break ties using one fixed total order on $\mathcal{A}^i$ at every state. The action-value vectors are identical within each orbit, so the selected greedy action is also identical within each orbit. This gives a G -invariant optimal stationary policy. Since this policy attains the unrestricted best-response value, equation 46 follows. $\square$

Lemma 29 (Nash-gap preservation). *Let M be a G -state-symmetric Markov game. Every quotient policy $\bar{\pi}$ satisfies*

$$\text{Nash gap}_M(\bar{\pi}^\uparrow) = \text{Nash gap}_{\bar{M}}(\bar{\pi}).$$

Consequently, $\bar{\pi}$ is $\in \text{Nash}$ in $\bar{M}$ if and only if $\bar{\pi}^\uparrow$ is $\in \text{Nash}$ in M .

Proof. Write $\pi = \bar{\pi}^\uparrow$. Fix an agent i . Since π^{-i} is invariant, Theorem 28 allows the original best-response optimization to be restricted to invariant local deviations:

$$\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) = \max_{\tilde{\pi}^i \text{ } G\text{-invariant}} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu).$$

By Theorem 26(a), projection is a bijection between invariant local policies on M and local policies on $\overline{M}$. Concretely, for an invariant deviation $\tilde{\pi}^i$, define

$$\bar{\nu}^i(a^i | \bar{s}) := \tilde{\pi}^i(a^i | \iota(\bar{s})).$$

Then $(\tilde{\pi}^i, \pi^{-i}) = (\bar{\nu}^i, \bar{\pi}^{-i})^\uparrow$. Value equivalence gives

$$\begin{aligned} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) &= \bar{V}_{\bar{\nu}^i, \bar{\pi}^{-i}}^i(\bar{\mu}), \\ V_{\pi}^i(\mu) &= \bar{V}_{\bar{\pi}}^i(\bar{\mu}). \end{aligned}$$

Therefore,

$$\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) = \max_{\bar{\nu}^i} \bar{V}_{\bar{\nu}^i, \bar{\pi}^{-i}}^i(\bar{\mu}) - \bar{V}_{\bar{\pi}}^i(\bar{\mu}).$$

The left and right sides are respectively agent i 's contributions to the Nash gaps in M and $\overline{M}$. Maximizing over i proves the identity in equation 8. $\square$

E BLACK-BOX RATE TRANSFER THROUGH THE QUOTIENT

Proof of Theorem 8, part 3. By hypothesis, the uniform MPG certificate applies to $\overline{M}$. Its state count is $|\overline{\mathcal{S}}|$, while the certificate parameters collected in ϑ are valid for both games. Applying equation 7 to $\overline{M}$ gives

$$\min_{0 \leq t < T} \text{Nash gap}_{\overline{M}}(\bar{\pi}_t) \leq \epsilon$$

whenever $T \geq T_{\text{MG}}(\epsilon, |\overline{\mathcal{S}}|; \vartheta)$. For every iterate, equation 8 gives

$$\text{Nash gap}_M(\pi_t) = \text{Nash gap}_M(\pi_t^\uparrow) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_t).$$

This proves equation 9.

For the displayed monomial certificate, applying the same sufficient-bound formula to M and $\overline{M}$ gives

$$\begin{aligned} B_{\text{full}}(\epsilon) &= C(\vartheta) |\mathcal{S}|^\alpha \epsilon^{-p}, \\ B_{\text{quot}}(\epsilon) &= C(\vartheta) |\overline{\mathcal{S}}|^\alpha \epsilon^{-p}. \end{aligned}$$

Taking their ratio proves equation 10. The argument uses only a convergence theorem for MPGs and does not invoke a single-agent convergence result.

If a particular external rate theorem also depends monotonically on visitation regularity, that theorem may additionally use Theorem 27(b) and (c). In particular, the original mismatch coefficient κ is a common valid upper bound because the quotient coefficient is at most κ . These secondary refinements are not needed for the state-count transfer stated in the main theorem. $\square$

F PERMUTATION INVARIANCE AND SHARED POINTED-STATE POLICIES

Lemma 30 (Pointed-state orbits and shared policies). *For pointed states (i, s) and (j, s') , the equality $z_i(s) = z_j(s')$ holds if and only if*

$$s^i = s'^j, \quad \{\{s^k : k \neq i\}\} = \{\{s'^k : k \neq j\}\}.$$

Moreover, a product policy is permutation invariant if and only if it is pointed-orbit shared.

Proof. If $z_i(s) = z_j(s')$, then, by definition of the orbit, some $\sigma \in S_N$ satisfies $j = \sigma(i)$ and $s' = \sigma \cdot s$. Then

$$s'^j = (\sigma \cdot s)^{\sigma(i)} = s^i,$$

and the coordinates of s' other than j are a reordering of the coordinates of s other than i . Hence the two concrete descriptors agree.

Conversely, suppose the two concrete descriptors agree. Set $\sigma(i) = j$. Equality of their multisets gives a bijection

$$\sigma : \mathcal{N} \setminus \{i\} \longrightarrow \mathcal{N} \setminus \{j\}$$

such that $s'^{\sigma(k)} = s^k$ for every $k \neq i$. Together with $s'^j = s^i$, this defines a permutation of $\mathcal{N}$ satisfying

$$(\sigma \cdot s)^{\sigma(k)} = s^k = s'^{\sigma(k)} \quad \text{for every } k \in \mathcal{N}.$$

Thus $\sigma \cdot s = s'$ and $\sigma \cdot (i, s) = (j, s')$. This proves the orbit characterization.

Now let π be permutation invariant. If $z_i(s) = z_j(s')$, choose σ as above. Then equation 14 gives

$$\pi^j(b \mid s') = \pi^{\sigma(i)}(b \mid \sigma \cdot s) = \pi^i(b \mid s).$$

Consequently,

$$\psi(b \mid z) := \pi^i(b \mid s) \quad \text{for any } (i, s) \text{ such that } z_i(s) = z$$

is well defined and gives equation 15.

Conversely, suppose equation 15 holds. For every σ ,

$$z_{\sigma(i)}(\sigma \cdot s) = z_i(s),$$

because the two pointed states are in the same orbit. Therefore

$$\pi^{\sigma(i)}(b \mid \sigma \cdot s) = \psi(b \mid z_{\sigma(i)}(\sigma \cdot s)) = \psi(b \mid z_i(s)) = \pi^i(b \mid s),$$

so π is permutation invariant.

Every $z \in \mathcal{Z}$ is the orbit of some pointed state. It follows that the shared rule inducing an invariant policy is unique, so the correspondence is a bijection. Finally, per-agent invariance implies the joint identity

$$\pi(\sigma \cdot a \mid \sigma \cdot s) = \pi(a \mid s), \tag{47}$$

because

$$\pi(\sigma \cdot a \mid \sigma \cdot s) = \prod_{i \in \mathcal{N}} \pi^{\sigma(i)}(a^i \mid \sigma \cdot s) = \prod_{i \in \mathcal{N}} \pi^i(a^i \mid s).$$

□

G VALUE, ADVANTAGE, VISITATION, AND POTENTIAL EQUIVARIANCE

Lemma 31 (Strategic equivariance under agent relabeling). *Under Theorem 10, every $\pi \in \Pi_{S_N}(M)$ satisfies, for every $\sigma \in S_N$, $i \in \mathcal{N}$, $s \in \mathcal{S}$, $a \in \mathcal{B}^N$, and $b \in \mathcal{B}$,*

$$\begin{aligned} V_{\pi}^{\sigma(i)}(\sigma \cdot s) &= V_{\pi}^i(s), & Q_{\pi}^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) &= Q_{\pi}^i(s, a), \\ A_{\pi}^{\sigma(i)}(\sigma \cdot s, b) &= A_{\pi}^i(s, b), & d_{\mu}^{\pi}(\sigma \cdot s) &= d_{\mu}^{\pi}(s), \\ \Phi_{\pi}(\sigma \cdot s) &= \Phi_{\pi}(s). \end{aligned} \tag{48}$$

In particular, there is a well-defined function $\hat{A}_{\pi}$ such that

$$A_{\pi}^i(s, b) = \hat{A}_{\pi}(z_i(s), b). \tag{49}$$

Proof. First consider the state kernel induced by π :

$$P^{\pi}(s' \mid s) := \sum_{a \in \mathcal{B}^N} \pi(a \mid s) P(s' \mid s, a).$$

Reindexing the action sum by $a' = \sigma \cdot a$ and using equation 47 and equation 11 gives

$$\begin{aligned} P^{\pi}(\sigma \cdot s' \mid \sigma \cdot s) &= \sum_{a \in \mathcal{B}^N} \pi(\sigma \cdot a \mid \sigma \cdot s) P(\sigma \cdot s' \mid \sigma \cdot s, \sigma \cdot a) \\ &= P^{\pi}(s' \mid s). \end{aligned} \tag{50}$$

The map

$$(s_t, a_t)_{t \geq 0} \mapsto (\sigma \cdot s_t, \sigma \cdot a_t)_{t \geq 0}$$

is therefore a probability-preserving bijection between trajectories under π started from s and trajectories under π started from $\sigma \cdot s$. By equation 12, the reward of agent $\sigma(i)$ along the transformed trajectory equals the reward of agent i along the original trajectory at every time. It follows that

$$V_{\pi}^{\sigma(i)}(\sigma \cdot s) = V_{\pi}^i(s).$$

Applying the same reindexing to the one-step action-value equations gives

$$Q_{\pi}^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = Q_{\pi}^i(s, a).$$

The same trajectory argument with the invariant signal ϕ in place of r^i proves

$$\Phi_{\pi}(\sigma \cdot s) = \Phi_{\pi}(s).$$

For completeness, let $p_t^{\pi}(s' | s)$ be the t -step transition probability under P^{π} . Equation (50) implies by induction that

$$p_t^{\pi}(\sigma \cdot s' | \sigma \cdot s) = p_t^{\pi}(s' | s) \quad \text{for every } t \geq 0.$$

Using equation 13 and reindexing the initial state as $s_0 = \sigma \cdot u$,

$$\begin{aligned} d_{\mu}^{\pi}(\sigma \cdot s') &= \sum_{t=0}^{\infty} \gamma^t \sum_{s_0 \in \mathcal{S}} \mu(s_0) p_t^{\pi}(\sigma \cdot s' | s_0) \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{u \in \mathcal{S}} \mu(\sigma \cdot u) p_t^{\pi}(\sigma \cdot s' | \sigma \cdot u) \\ &= d_{\mu}^{\pi}(s'). \end{aligned}$$

Subtracting the value identity from the action-value identity gives joint advantage equivariance:

$$A_{\pi}^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = A_{\pi}^i(s, a). \quad (51)$$

Fix $b \in \mathcal{B}$ and reindex the other agents' actions by $a' = \sigma \cdot a$ with $a^i = b$. Since $(\sigma \cdot a)^{\sigma(i)} = b$, product-policy invariance gives

$$\pi^{-\sigma(i)}\left((\sigma \cdot a)^{-\sigma(i)} \mid \sigma \cdot s\right) = \pi^{-i}(a^{-i} \mid s).$$

Applying this identity and equation 51 in the definition equation 1 yields

$$A_{\pi}^{\sigma(i)}(\sigma \cdot s, b) = A_{\pi}^i(s, b).$$

Finally, if $z_i(s) = z_j(s')$, then Theorem 30 supplies a σ with $j = \sigma(i)$ and $s' = \sigma \cdot s$. The local-advantage identity therefore gives

$$A_{\pi}^j(s', b) = A_{\pi}^i(s, b).$$

Thus defining $\hat{A}_{\pi}(z, b)$ using any pointed state with descriptor z is independent of the choice and proves equation 49. $\square$

H POINTED-ORBIT BEST RESPONSES AND THE UNRESTRICTED NASH GAP

Lemma 32 (Pointed-orbit optimal best response). *Under Theorem 10, let $\pi \in \Pi_{\mathcal{S}_N}(M)$. For every agent i , the best-response identity equation 18 holds. Moreover, all agents have the same best-response gain, and hence equation 19 holds.*

Proof. Fix an agent i and let

$$H_i := \{\tau \in S_N : \tau(i) = i\}$$

be its stabilizer subgroup. Against π^{-i} , agent i faces the finite discounted MDP with state space $\mathcal{S}$, action space $\mathcal{B}$, and

$$\begin{aligned} P_{\pi^{-i}}^i(s' | s, b) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) P(s' | s, b, a^{-i}), \\ r_{\pi^{-i}}^i(s, b) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) r^i(s, b, a^{-i}). \end{aligned}$$

For $\tau \in H_i$, reindex the sum over a^{-i} by the permutation of the other agents' action coordinates induced by τ . Because $\tau(i) = i$, the focal action b is unchanged. Permutation invariance of π gives

$$\pi^{-i}((\tau \cdot a)^{-i} | \tau \cdot s) = \pi^{-i}(a^{-i} | s) \quad \text{when } a^i = b.$$

Using equation 11 and equation 12, with $\tau(i) = i$, therefore gives

$$P_{\pi^{-i}}^i(\tau \cdot s' | \tau \cdot s, b) = P_{\pi^{-i}}^i(s' | s, b), \quad (52)$$

$$r_{\pi^{-i}}^i(\tau \cdot s, b) = r_{\pi^{-i}}^i(s, b). \quad (53)$$

Thus the best-response MDP is H_i -state-symmetric with unchanged action labels.

Its Bellman optimality operator maps H_i -invariant value functions to H_i -invariant value functions by equation 52–equation 53. Starting value iteration from zero and using contraction therefore shows that the optimal value and optimal action-value functions are H_i invariant. Choosing a greedy action with one fixed tie-breaking order on $\mathcal{B}$ gives an H_i -invariant optimal stationary policy.

The H_i -orbits in $\mathcal{S}$ are exactly the fibers of z_i . Indeed, elements of H_i fix s^i and reorder the other coordinates; conversely, equal descriptors allow the remaining equal-valued coordinates to be matched by a permutation fixing i . Hence an H_i -invariant local policy is exactly a policy of the form $\chi \circ z_i$. The invariant optimal policy constructed above therefore proves equation 18.

It remains to compare agents. Fix i, j and choose $\sigma \in S_N$ with $\sigma(i) = j$. For any local policy ν^i , define its relabeling for agent j by

$$(\sigma_* \nu^i)(b | s') := \nu^i(b | \sigma^{-1} \cdot s'). \quad (54)$$

This is a bijection between the local policy spaces of agents i and j . Let

$$q := (\nu^i, \pi^{-i}), \quad q' := (\sigma_* \nu^i, \pi^{-j}).$$

For the deviating coordinate,

$$q'^{\sigma(i)}(b | \sigma \cdot s) = q^i(b | s)$$

by equation 54; for every $k \neq i$, the same identity holds between $q'^{\sigma(k)}$ and q^k by permutation invariance of π . Thus the trajectory relabeling used in Theorem 31, now between policies q and q' , gives

$$V_{q'}^j(\sigma \cdot s) = V_q^i(s).$$

Averaging over the permutation-invariant initial distribution and reindexing states yields

$$V_{\sigma_* \nu^i, \pi^{-j}}^j(\mu) = V_{\nu^i, \pi^{-i}}^i(\mu).$$

Taking maxima and using the bijectivity of $\nu^i \mapsto \sigma_* \nu^i$ shows that the optimal response values are equal for agents i and j . Similarly, equation 48 and equation 13 imply

$$V_{\pi}^j(\mu) = V_{\pi}^i(\mu).$$

The best-response gains are therefore equal. Combining this fact with equation 18 and the definition of the Nash gap proves equation 19. $\square$

I BLACK-BOX RATE TRANSFER FOR SHARED RULES

Proof of Theorem 12, part 3. Every solver iterate is pointed-orbit shared by hypothesis. Equation (19) therefore gives, for every fixed agent i and every iteration t ,

$$\text{Nash gap}_M(\pi_t) = \max_{\chi: \mathcal{Z} \rightarrow \Delta(\mathcal{B})} \left[V_{\chi \circ z_i, \pi_t^{-i}}^i(\mu) - V_{\pi_t}^i(\mu) \right].$$

Taking the minimum over $0 \leq t < T$ and applying equation 17 proves equation 20.

For the optional bound comparison, the unrestricted table has $D_{\text{full}} = N |\mathcal{C}|^N$ local policy blocks, while the shared table has $D_{\text{perm}} = |\mathcal{Z}|$. Orbit compatibility is precisely the additional hypothesis that the same certificate form controls the ordinary Nash gap before tying and the pointed-orbit gap after tying, using common auxiliary bounds. Both representations have state count $S = |\mathcal{C}|^N$. Therefore

$$\frac{B_{\text{perm}}(\epsilon)}{B_{\text{full}}(\epsilon)} = \frac{C_{\text{perm}}(\vartheta) S^\alpha D_{\text{perm}}^\delta \epsilon^{-p}}{C_{\text{full}}(\vartheta) S^\alpha D_{\text{full}}^\delta \epsilon^{-p}} \leq \left(\frac{|\mathcal{Z}|}{N |\mathcal{C}|^N} \right)^\delta.$$

This proves equation 21. The condition on ϑ is essential: for Euclidean shared-coordinate gradient methods, for example, smoothness may depend on the largest number of pointed states tied to one descriptor. The block count alone does not control that multiplicity. $\square$

J PROOF OF THE COMBINED REDUCTION

Proof of Theorem 16. The descended permutation action. Suppose $s, \tilde{s} \in \bar{s}$, so $\tilde{s} = L_g[s]$ for some $g \in G$. By equation 22,

$$\sigma \cdot \tilde{s} = \sigma \cdot L_g[s] = L_g[\sigma \cdot s].$$

Thus $\sigma \cdot \tilde{s}$ and $\sigma \cdot s$ belong to the same G orbit, so $\sigma \cdot \bar{s} := \rho(\sigma \cdot s)$ is independent of the representative. It is a group action because the original S_N action is one.

For quotient states $\bar{s}, \bar{s}'$ with representatives s, s' and any joint action a , reindex the target orbit by $u = \sigma \cdot v$ and use equation 11:

$$\begin{aligned} \bar{P}(\sigma \cdot \bar{s}' \mid \sigma \cdot \bar{s}, \sigma \cdot a) &= \sum_{u \in \sigma \cdot \bar{s}'} P(u \mid \sigma \cdot s, \sigma \cdot a) \\ &= \sum_{v \in \bar{s}'} P(\sigma \cdot v \mid \sigma \cdot s, \sigma \cdot a) \\ &= \sum_{v \in \bar{s}'} P(v \mid s, a) = \bar{P}(\bar{s}' \mid \bar{s}, a). \end{aligned}$$

The reward and potential identities follow directly:

$$\begin{aligned} \bar{r}^{\sigma(i)}(\sigma \cdot \bar{s}, \sigma \cdot a) &= r^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = r^i(s, a) = \bar{r}^i(\bar{s}, a), \\ \bar{\phi}(\sigma \cdot \bar{s}, \sigma \cdot a) &= \phi(\sigma \cdot s, \sigma \cdot a) = \phi(s, a) = \bar{\phi}(\bar{s}, a). \end{aligned}$$

Finally, reindexing the orbit sum and using equation 13 gives

$$\bar{\mu}(\sigma \cdot \bar{s}) = \sum_{u \in \sigma \cdot \bar{s}} \mu(u) = \sum_{v \in \bar{s}} \mu(\sigma \cdot v) = \bar{\mu}(\bar{s}).$$

Hence $\bar{M}$ is permutation symmetric.

The two orbit orders agree. Commutativity makes the displayed formula in equation 23 a left action of $G \times S_N$. Define

$$\begin{aligned} F: \mathcal{W} &\longrightarrow (\mathcal{N} \times (\mathcal{S}/G))/S_N, \\ F([(i, s)]_{G \times S_N}) &:= [(i, \rho(s))]_{S_N}. \end{aligned}$$

The representative calculation above shows that F is well defined. It is surjective because every pointed quotient state has a representative in $\mathcal{N} \times \mathcal{S}$. If $F([(i, s)]) = F([(j, s')])$, there is a $\sigma \in S_N$ with $j = \sigma(i)$ and $\rho(s') = \rho(\sigma \cdot s)$. Therefore $s' = L_g[\sigma \cdot s]$ for some $g \in G$, so the two pointed states are in the same $G \times S_N$ orbit. Thus F is injective.

Similarly, commutativity induces a well-defined G action on $\mathcal{Z} = (\mathcal{N} \times \mathcal{S})/S_N$ by

$$g \cdot [(i, s)]_{S_N} := [(i, L_g[s])]_{S_N}.$$

Quotienting this orbit space by G identifies exactly the same pairs of pointed states as the product action. This proves both isomorphisms in equation 24.

Jointly invariant policies. If π is jointly invariant and $(j, s') = (\sigma(i), L_g[\sigma \cdot s])$, then

$$\begin{aligned} \pi^j(b \mid s') &= \pi^{\sigma(i)}(b \mid L_g[\sigma \cdot s]) \\ &= \pi^{\sigma(i)}(b \mid \sigma \cdot s) = \pi^i(b \mid s). \end{aligned}$$

Thus the local policy is constant on every combined pointed orbit and defines one rule on $\mathcal{W}$. Conversely, a rule on $\mathcal{W}$ is unchanged under the elements (g, e) and (e, σ) , so its induced policy is both G invariant and permutation invariant. This proves the policy correspondence.

Unrestricted best responses. Fix an agent i and let $H_i := \{\tau \in S_N : \tau(i) = i\}$. Against jointly invariant opponents, agent i 's best-response MDP is symmetric under $G \times H_i$. Indeed, reindexing the opponents' action sum by τ and applying first the permutation identities and then the state-only identities gives, for every $g \in G$, $\tau \in H_i$, and focal action b ,

$$\begin{aligned} P_{\pi-i}^i(L_g[\tau \cdot s'] \mid L_g[\tau \cdot s], b) &= P_{\pi-i}^i(s' \mid s, b), \\ r_{\pi-i}^i(L_g[\tau \cdot s], b) &= r_{\pi-i}^i(s, b). \end{aligned}$$

The focal action remains b because G fixes all action labels and $\tau(i) = i$. Value iteration from zero therefore yields a $G \times H_i$ -invariant optimal value and action-value function. Greedy selection with one fixed tie-breaking order gives a $G \times H_i$ -invariant optimal stationary response.

The $G \times H_i$ orbits of states are exactly the fibers of w_i . One implication follows immediately from equation 23. For the converse, if $w_i(s) = w_i(s')$, then some (g, σ) maps (i, s) to (i, s') ; hence $\sigma(i) = i$, so $\sigma \in H_i$ and $s' = L_g[\sigma \cdot s]$. The invariant optimal response is therefore of the form $\chi \circ w_i$. Finally, the relabeling argument in Theorem 32 still applies because a jointly invariant policy is permutation invariant. It shows that all agents have the same best-response gain. This proves equation 25.

Combined rate transfer. By the first part, the shared solver may be run on $\overline{M}$. Its pointed-state orbit space is identified with $\mathcal{W}$ by equation 24. Each iterate $\bar{\pi}_t$ is permutation invariant. For a fixed agent i , its best-response MDP is H_i symmetric, where $H_i = \{\tau \in S_N : \tau(i) = i\}$, by the same calculation as in Equations (52) and (53). The H_i orbits of $\overline{\mathcal{S}}$ are exactly the fibers of $z_i^{\overline{M}}$, and relabeling agents shows that all best-response gains are equal. Therefore

$$\text{OrbGap}_{\overline{M}}^i(\bar{\pi}_t) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_t).$$

Lifting and equation 8 then give, at every iteration,

$$\text{Nash gap}_M(\bar{\pi}_t^\uparrow) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_t).$$

Applying the assumed shared-solver certificate with $S = |\overline{\mathcal{S}}|$ and $D = |\mathcal{W}|$ proves equation 26.

Under the orbit-compatible monomial certificate, the full table has state count $|\mathcal{S}|$ and $N|\mathcal{S}|$ policy blocks, the permutation-only representation has state count $|\mathcal{S}|$ and $|\mathcal{Z}|$ blocks, the quotient-only representation has state count $|\overline{\mathcal{S}}|$ and $N|\overline{\mathcal{S}}|$ blocks, and the combined representation has state

count $|\overline{\mathcal{S}}|$ and $|\mathcal{W}|$ blocks. The assumed prefactor inequalities therefore give

$$\begin{aligned}\frac{B_{\text{both}}(\epsilon)}{B_{\text{full}}(\epsilon)} &\leq \left(\frac{|\overline{\mathcal{S}}|}{|\mathcal{S}|}\right)^\alpha \left(\frac{|\mathcal{W}|}{N|\mathcal{S}|}\right)^\delta \\ &= \left(\frac{|\overline{\mathcal{S}}|}{|\mathcal{S}|}\right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|}\right)^\delta \left(\frac{|\mathcal{Z}|}{N|\mathcal{S}|}\right)^\delta, \\ \frac{B_{\text{both}}(\epsilon)}{B_{\text{perm}}(\epsilon)} &\leq \left(\frac{|\overline{\mathcal{S}}|}{|\mathcal{S}|}\right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|}\right)^\delta, \\ \frac{B_{\text{both}}(\epsilon)}{B_{\text{quot}}(\epsilon)} &\leq \left(\frac{|\mathcal{W}|}{N|\overline{\mathcal{S}}|}\right)^\delta.\end{aligned}$$

This proves Equations (27) to (29). The orbit isomorphisms identify $\mathcal{W}$ with $\mathcal{Z}/G$, so $|\mathcal{W}| \leq |\mathcal{Z}|$. Also, $\mathcal{W}$ is an orbit space of $\mathcal{N} \times \overline{\mathcal{S}}$, so $|\mathcal{W}| \leq N|\overline{\mathcal{S}}|$. If both stated G actions are free, the orbit-stabilizer theorem gives

$$|\overline{\mathcal{S}}| = \frac{|\mathcal{S}|}{|G|}, \quad |\mathcal{W}| = \frac{|\mathcal{Z}|}{|G|}.$$

These are separate assumptions: freeness on $\mathcal{S}$ need not imply freeness of the induced action on $\mathcal{Z}$, because $L_g[s]$ may agree with s after an agent relabeling. Therefore both are needed for the displayed multiplicative factors. The two equalities above show that $|\overline{\mathcal{S}}|/|\mathcal{S}|$ and $|\mathcal{W}|/|\mathcal{Z}|$ both equal $|G|^{-1}$. Substitution proves Equations (30) and (31). $\square$

K PROOFS OF THE SAMPLE-COMPLEXITY RESULTS

This section proves Theorems 17 to 19. The first three lemmas are stated for a generic game so that all three results can share them. Throughout, X denotes a finite MPG $\langle \mathcal{N}, \mathcal{S}_X, \mathcal{A}_X, P_X, \vec{r}_X, \mu_X, \gamma \rangle$ with potential ϕ_X whose total potential Φ_π^X takes values in $[\Phi_{\min}, \Phi_{\max}]$ for every stationary product policy π and state x . By Theorem 3 this holds for $X = M$ with $\Phi_\pi^M = \Phi_\pi$, and by Theorem 27(a) for $X = \overline{M}$ with $\Phi_\pi^{\overline{M}} = \overline{\Phi}_\pi$.

The *potential MDP* of X has state space $\mathcal{S}_X$, action space $\mathcal{A}_X$, reward ϕ_X , kernel P_X , and discount γ ; the value it assigns to a stationary product policy π is the total potential Φ_π^X . Let

$$\Phi_\star^X(x) := \max_\pi \Phi_\pi^X(x)$$

denote its optimal value, the maximum ranging over all policies of that MDP, including history-dependent ones, and write $\Phi_\star^X(\mu_X) := \mathbb{E}_{x \sim \mu_X}[\Phi_\star^X(x)]$. A finite discounted MDP admits a stationary deterministic optimal policy, obtained by greedy selection with respect to its optimal action values; fix such a policy $\pi_\star$ for the potential MDP of X . For a transition kernel Q on $\mathcal{S}_X \times \mathcal{A}_X$, let Φ_π^Q denote the value functions of the potential MDP with Q in place of P_X , fix a stationary deterministic policy π_Q optimal for it, and set

$$\varepsilon_Q := \max_{(x,a) \in \mathcal{S}_X \times \mathcal{A}_X} \|Q(\cdot | x, a) - P_X(\cdot | x, a)\|_1, \quad \eta_Q := \frac{\gamma \varepsilon_Q \Phi_{\sup}^X}{1 - \gamma}.$$

Stationary deterministic policies of the potential MDP select one joint action per state and are therefore product policies of X .

Lemma 33 (Simulation lemma for the potential MDP). *Every stationary product policy π of X satisfies*

$$\|\Phi_\pi^Q - \Phi_\pi^X\|_\infty \leq \eta_Q.$$

Proof. For a stationary policy π of the potential MDP, define the averaged reward and the averaged kernel operators

$$\begin{aligned}\phi_{X,\pi}(x) &:= \sum_{a \in \mathcal{A}_X} \pi(a | x) \phi_X(x, a), \\ (P_{X,\pi}v)(x) &:= \sum_{a \in \mathcal{A}_X} \pi(a | x) \sum_{x' \in \mathcal{S}_X} P_X(x' | x, a) v(x'),\end{aligned}$$

with Q_π defined from Q in the same way. The value functions are the unique fixed points of the Bellman equations

$$\begin{aligned}\Phi_\pi^X &= \phi_{X,\pi} + \gamma P_{X,\pi} \Phi_\pi^X, \\ \Phi_\pi^Q &= \phi_{X,\pi} + \gamma Q_\pi \Phi_\pi^Q.\end{aligned}$$

Subtracting,

$$\Phi_\pi^Q - \Phi_\pi^X = \gamma Q_\pi (\Phi_\pi^Q - \Phi_\pi^X) + \gamma (Q_\pi - P_{X,\pi}) \Phi_\pi^X.$$

The averaged operator satisfies $\|Q_\pi v\|_\infty \leq \|v\|_\infty$, and Hölder's inequality applied statewise after averaging over actions gives

$$\|(Q_\pi - P_{X,\pi})v\|_\infty \leq \varepsilon_Q \|v\|_\infty.$$

Taking supremum norms and rearranging,

$$\|\Phi_\pi^Q - \Phi_\pi^X\|_\infty \leq \frac{\gamma \varepsilon_Q}{1 - \gamma} \|\Phi_\pi^X\|_\infty.$$

By hypothesis, the total potential of every product policy takes values in $[\Phi_{\min}, \Phi_{\max}]$, so $\|\Phi_\pi^X\|_\infty \leq \Phi_{\sup}$. $\square$

Lemma 34 (Certainty-equivalent optimality transfer). *For every $x \in \mathcal{S}_X$,*

$$\Phi_{\pi_Q}^X(x) \geq \Phi_\star^X(x) - 2\eta_Q.$$

Proof. The policies π_Q and $\pi_\star$ are stationary deterministic joint policies, hence product policies, so Theorem 33 applies to both. For every x ,

$$\begin{aligned}\Phi_{\pi_Q}^X(x) &\geq \Phi_{\pi_Q}^Q(x) - \eta_Q \\ &\geq \Phi_{\pi_\star}^Q(x) - \eta_Q \\ &\geq \Phi_{\pi_\star}^X(x) - 2\eta_Q = \Phi_\star^X(x) - 2\eta_Q.\end{aligned}$$

The middle inequality holds because π_Q is optimal for the Q -model potential MDP, whose optimal value dominates the Q -model value of every policy, in particular of $\pi_\star$. $\square$

Lemma 35 (Nash gap of an approximate potential maximizer). *Let a stationary product policy π of X satisfy $\Phi_\pi^X(x) \geq \Phi_\star^X(x) - \beta$ for every $x \in \mathcal{S}_X$ and some $\beta \geq 0$. Then $\text{Nash gap}_X(\pi) \leq \beta$.*

Proof. Fix an agent i and a stationary local deviation ν^i . The deviated policy (ν^i, π^{-i}) is a stationary product policy and hence also a policy of the potential MDP of X , so its total potential is dominated by the optimal value:

$$\Phi_{\nu^i, \pi^{-i}}^X(x) \leq \Phi_\star^X(x) \quad \text{for every } x \in \mathcal{S}_X.$$

Averaging over $x \sim \mu_X$ and applying the MPG identity of X at μ_X ,

$$\begin{aligned}V_{\nu^i, \pi^{-i}}^i(\mu_X) - V_\pi^i(\mu_X) &= \Phi_{\nu^i, \pi^{-i}}^X(\mu_X) - \Phi_\pi^X(\mu_X) \\ &\leq \Phi_\star^X(\mu_X) - (\Phi_\star^X(\mu_X) - \beta) = \beta.\end{aligned}$$

Maximizing over ν^i and over agents bounds the Nash gap by β . $\square$

Lemma 36 (Uniform ℓ_1 concentration of the pooled estimator). *Let M be a G -state-symmetric MPG under Theorems 3 and 5, and let $\hat{P}$ be the estimator equation 32 with per-pair budget n . Fix $\varepsilon_P > 0$ and $\delta \in (0, 1)$. If*

$$n \geq \frac{2}{\varepsilon_P^2} \left(|\bar{\mathcal{S}}| \log 2 + \log \frac{|\bar{\mathcal{S}}| |\mathcal{A}|}{\delta} \right),$$

then, with probability at least $1 - \delta$,

$$\max_{(\bar{s}, a) \in \bar{\mathcal{S}} \times \mathcal{A}} \left\| \hat{P}(\cdot \mid \bar{s}, a) - \bar{P}(\cdot \mid \bar{s}, a) \right\|_1 \leq \varepsilon_P. \quad (55)$$

Proof. Fix a pair $(\bar{s}, a)$ and abbreviate $p := \bar{P}(\cdot \mid \bar{s}, a)$ and $\hat{p} := \hat{P}(\cdot \mid \bar{s}, a)$. For a draw $s' \sim P(\cdot \mid \iota(\bar{s}), a)$ and any subset $B \subseteq \bar{\mathcal{S}}$,

$$\Pr[\rho(s') \in B] = \sum_{\bar{s}' \in B} \sum_{u \in \bar{s}'} P(u \mid \iota(\bar{s}), a) = p(B), \quad (56)$$

so $\hat{p}(B)$ is the empirical mean of n independent Bernoulli variables with mean $p(B)$. For any two probability vectors on $\bar{\mathcal{S}}$,

$$\|\hat{p} - p\|_1 = 2 \max_{B \subseteq \bar{\mathcal{S}}} (\hat{p}(B) - p(B)),$$

the maximum being attained at $B = \{\bar{s}' : \hat{p}(\bar{s}') > p(\bar{s}')\}$. Hoeffding's inequality gives, for each fixed B ,

$$\Pr \left[\hat{p}(B) - p(B) \geq \frac{\varepsilon_P}{2} \right] \leq \exp \left(-\frac{n\varepsilon_P^2}{2} \right).$$

A union bound over the $2^{|\bar{\mathcal{S}}|}$ subsets, and then over the $|\bar{\mathcal{S}}| |\mathcal{A}|$ quotient pairs, bounds the failure probability of equation 55 by

$$|\bar{\mathcal{S}}| |\mathcal{A}| 2^{|\bar{\mathcal{S}}|} \exp \left(-\frac{n\varepsilon_P^2}{2} \right) \leq \delta$$

under the stated budget. $\square$

Lemma 37 (Uniform ℓ_1 concentration of the transported estimator). *Let X additionally be permutation symmetric in the sense of the shared-solver rate convention, and let $\tilde{P}$ be the transported estimator equation 37 with per-orbit budget n . Fix $\varepsilon_P > 0$ and $\delta \in (0, 1)$. If*

$$n \geq \frac{2}{\varepsilon_P^2} \left(|\mathcal{S}_X| \log 2 + \log \frac{|\mathcal{Y}_X|}{\delta} \right),$$

then, with probability at least $1 - \delta$,

$$\max_{(x, a) \in \mathcal{S}_X \times \mathcal{B}_X^N} \left\| \tilde{P}(\cdot \mid x, a) - P_X(\cdot \mid x, a) \right\|_1 \leq \varepsilon_P. \quad (57)$$

Proof. Fix a pair (x, a) , write $\bar{y} := y(x, a)$, $(x_0, a_0) := j(\bar{y})$, and $\tau := \tau_{x, a}$, so that $\tau \cdot (x_0, a_0) = (x, a)$. The transition symmetry identity of the permutation-symmetric class, applied with the permutation τ , states that $P_X(\tau \cdot u \mid \tau \cdot x_0, \tau \cdot a_0) = P_X(u \mid x_0, a_0)$ for every u . Substituting $u = \tau^{-1} \cdot x'$ gives the row identity

$$P_X(x' \mid x, a) = P_X(\tau^{-1} \cdot x' \mid x_0, a_0). \quad (58)$$

Combining equation 37 and equation 58, and reindexing the sum over x' by the bijection $x' \mapsto \tau^{-1} \cdot x'$,

$$\begin{aligned} \left\| \tilde{P}(\cdot \mid x, a) - P_X(\cdot \mid x, a) \right\|_1 &= \sum_{x' \in \mathcal{S}_X} \left| \hat{P}_X(\tau^{-1} \cdot x' \mid x_0, a_0) - P_X(\tau^{-1} \cdot x' \mid x_0, a_0) \right| \\ &= \left\| \hat{P}_X(\cdot \mid x_0, a_0) - P_X(\cdot \mid x_0, a_0) \right\|_1. \end{aligned}$$

Thus equation 57 holds as soon as every representative row is ε_P -accurate in ℓ_1 . For a fixed representative row and any subset $B \subseteq \mathcal{S}_X$, the empirical mass $\hat{P}_X(B \mid x_0, a_0)$ is the empirical mean of n independent Bernoulli variables with mean $P_X(B \mid x_0, a_0)$, so, exactly as in the proof of Theorem 36, Hoeffding's inequality and a union bound over the $2^{|\mathcal{S}_X|}$ subsets and the $|\mathcal{Y}_X|$ representative rows bound the failure probability by

$$|\mathcal{Y}_X| 2^{|\mathcal{S}_X|} \exp\left(-\frac{n\varepsilon_P^2}{2}\right) \leq \delta$$

under the stated budget. $\square$

Proof of Theorem 17. Instantiate $X := \overline{M}$ and $Q := \widehat{P}$, so that $\pi_Q = \bar{\pi}_{ce}$; by Theorem 27, X satisfies the standing hypotheses of this section. Set

$$\varepsilon_P := \begin{cases} \frac{(1-\gamma)\epsilon}{2\gamma\Phi_{\sup}}, & \gamma\Phi_{\sup} > 0, \\ 2, & \gamma\Phi_{\sup} = 0. \end{cases}$$

In the second case, equation 55 holds surely because the ℓ_1 distance of two probability vectors never exceeds 2, and $\eta_Q \leq \gamma\varepsilon_P\Phi_{\sup}/(1-\gamma) = 0$; the budget equation 33 is also vacuous. In the first case, the budget equation 33 equals the budget of Theorem 36 at this ε_P , so equation 55 holds with probability at least $1 - \delta$, and on that event $\varepsilon_Q \leq \varepsilon_P$, hence $2\eta_Q \leq \epsilon$. In both cases, on the event equation 55, Theorems 34 and 35 give

$$\text{Nash gap}_{\overline{M}}(\bar{\pi}_{ce}) \leq 2\eta_Q \leq \epsilon.$$

By equation 8 in Theorem 8,

$$\text{Nash gap}_M(\bar{\pi}_{ce}^\uparrow) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_{ce}) \leq \epsilon,$$

which proves equation 34. Multiplying the per-pair budget equation 33 by the $|\overline{\mathcal{S}}| |\mathcal{A}|$ queried pairs gives the sufficient total budget $B^{\text{gen}}(\epsilon, \delta; |\overline{\mathcal{S}}|, |\mathcal{A}|)$ in equation 35.

For part 2., apply the entire argument with the trivial group $G = \{e\}$: every orbit is a singleton, the quotient is M itself, Theorem 4 and Theorem 5 hold vacuously, and equation 32 reduces to the standard empirical kernel at each pair (s, a) . The same pipeline run directly on M therefore has the sufficient budget $B^{\text{gen}}(\epsilon, \delta; |\mathcal{S}|, |\mathcal{A}|)$. The equality in equation 36 is direct substitution into equation 35. The inequality follows because $|\overline{\mathcal{S}}| \leq |\mathcal{S}|$ and, at fixed action count A , the map $S \mapsto S \log 2 + \log(SA/\delta)$ is nondecreasing for $S \geq 1$, so the second factor is at most one. $\square$

Proof of Theorem 18. Take $Q := \tilde{P}$, so that $\pi_Q = \pi_{ce}$, and choose ε_P exactly as in the proof of Theorem 17; the same case analysis applies, with Theorem 37 in place of Theorem 36 and the budget equation 38 in place of equation 33. On the event equation 57, $\varepsilon_Q \leq \varepsilon_P$ and $2\eta_Q \leq \epsilon$, so Theorems 34 and 35 give

$$\text{Nash gap}_X(\pi_{ce}) \leq 2\eta_Q \leq \epsilon.$$

Multiplying the per-orbit budget equation 38 by the $|\mathcal{Y}_X|$ queried orbits gives equation 39.

For part 2., the untied construction is Theorem 172. applied to X , with state count $|\mathcal{S}_X|$ and action count $|\mathcal{B}_X|^N$. The equality in equation 40 is direct substitution into equation 35 and equation 39. The inequality follows because the orbit map $(x, a) \mapsto y(x, a)$ is onto, so $|\mathcal{Y}_X| \leq |\mathcal{S}_X| |\mathcal{B}_X|^N$ and the second factor is at most one. $\square$

Proof of Theorem 19. By Theorem 16, $\overline{M}$ is permutation symmetric under the induced action $\sigma \cdot \bar{s} = \rho(\sigma \cdot s)$, and by Theorem 27 it satisfies Theorem 3 with the same total-potential bounds, hence the same $\Phi_{\sup}$. Its generative model is realized through M : querying a representative quotient pair means sampling $s' \sim P(\cdot \mid \iota(\bar{s}), a)$ and recording the orbit $\rho(s')$, which by equation 56 is an exact draw from $\bar{P}(\cdot \mid \bar{s}, a)$. Theorem 18 applied to $X = \overline{M}$, whose next-state space is $\bar{\mathcal{S}}$ and whose pair-orbit space is $\mathcal{Y}_{\overline{M}}$, then gives $\text{Nash gap}_{\overline{M}}(\bar{\pi}_{ce}) \leq \epsilon$ with probability at least $1 - \delta$ under the stated per-orbit budget, with total budget $n |\mathcal{Y}_{\overline{M}}|$. Finally, equation 8 transfers the bound: $\text{Nash gap}_M(\bar{\pi}_{ce}^\uparrow) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_{ce}) \leq \epsilon$. $\square$